5 Essentials of minimally invasive operative dentistry

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5.1 The oral healthcare team

All members of the oral healthcare team have a part to play in patient management, and the team is comprised of the lead dentist (plus other colleagues in the dental practice), the dental nurse, hygienist, receptionist, laboratory technician, and possibly a dental therapist. In the UK, registered dental nurses can take further qualifications in teaching, oral health education, and radiography, and can specialize in other aspects of dentistry, including orthodontics, oral surgery, sedation, and special care.

If the dentist wishes to have a second specialist opinion regarding a difficult diagnosis, formulating a care plan or even executing it, they may refer the patient to a specialist dentist working in another practice, or to a hospital-based consultant specialist in restorative dentistry. These specialists have undergone further postgraduate clinical and academic training and gained qualifications enabling them to be registered as specialists with the General Dental Council (GDC) in the UK in their specific trained fields (e.g. endodontics, periodontics, prosthodontics), or have further specialist training in restorative dentistry. The lead dentist will act as a central hub in the coordinating wheel of patient management, possibly outsourcing different aspects of work to relevant specialist colleagues, as spokes of that wheel.

5.2 The dental surgery or 'dental clinic'

This is the clinical environment in which patients are diagnosed and treated. This room has traditionally been known as the 'dental surgery', but a more appropriate modern description might be the 'dental clinic', as much of the more holistic care offered to patients within its four walls will be non-surgical in the first instance. The operator and nurse must work closely together. To be successful, each must build up an understanding of how the other works. The clinic consists of a dental operating chair with an attached or mobile bracket table carrying the rotary instruments and 3-1 air/water syringe (and possibly the light-cure unit and ultrasonic scaler), work surfaces (which should be as clutter-free as possible for good-quality infection control; see later), cupboards for storage, and two sinks, one for normal hand washing and another for decontaminating soiled instruments prior to sterilization. Often the surgery will also house an X-ray unit for taking intra-oral radiographs. Most clinics are designed to accommodate right-handed practitioners, in terms of the location of many of the instruments and controls. In larger clinics there may be space to accommodate a table/desk and comfortable chair where the initial verbal consultation may take place before moving to the dental chair for the clinical examination.

5.2.1 Positioning the dentist, patient, and nurse

All three must be positioned for maximum comfort, visibility, and access while maintaining a healthy posture. Initially, the dentist and patient should be sitting at the same eye level, face to face (see Figure 5.1).

The patient is then reclined in the dental chair into a near horizontal position, and the height of the chair adjusted so that the patient's head is level with the dentist's mid-sternum, whose knees are placed beneath the patient's headrest. The dentist should sit with their back straight and upper arms vertical, with the elbows bent at right angles. Their thighs should be near parallel to the floor and knees bent at right angles, with feet firmly placed flat on the floor (see Figure 5.2). The use of backless saddle chairs can help the clinician to achieve and maintain this posture throughout the working day.

For a right-handed dentist, the nurse should sit to the left of the patient, facing the patient, sitting approximately 10–15 cm higher than the dentist to aid their direct vision into the oral cavity. The nurse's chair may have a foot bar and a swivel back rest that can be moved round to support the nurse when they are assisting the dentist intra-orally. If the patient's head in the dental chair is represented by a clockface with 12 o'clock between the patient's eyes, the dentist can move between 8 o'clock (to view the lower right quadrant with direct vision and the patient slightly more upright in the chair) and 1 o'clock, and the nurse from 1 to 4 o'clock positions (see Figure 5.3). These positions may be reversed for the left-handed operator, assuming that the surgery is designed with this in mind. Some dental chairs are designed so that the nurse's suction and operator's bracket table/mobile cart may be transposed.

The patient can facilitate direct intra-oral vision by moving their head. The dentist should ask and gently guide the movement—a right head turn to view the upper left buccal quadrant, tipping the head down to view the mandibular dentition, and lifting the chin up to view the maxillary dentition.



Figure 5.1 Dentist and patient positioning during the consultation. Note how the dentist is sitting slightly in front of the patient and at the same eye level. This positioning enables the patient to feel more comfortable and relaxed.



Figure 5.2 Dentist operating position with the patient reclined in the dental chair. Note how the operator's elbows are bent at 90° and their back is straight and shoulders relaxed. The dentist is using an LED headlight to help to illuminate his field of view, without shadows.

5.2.2 Lighting

Good-quality illumination is vital in order to work in a patient's mouth. It is usually afforded by the overhead chair light fitted with daylight-simulating light bulbs. These lights may be focused on the patient's face and mouth by correct spatial positioning and alignment (some cast a horizontal dark shadow if incorrectly focused). They should be aligned to cast their beam of light as directly in line with the operator's line of sight as possible (usually offset by only a few degrees), to reduce the shadows cast intra-orally. The intra-oral mirror should be used to direct light specifically to the areas required (e.g. maxillary structures). Light handles must be covered with disposable shields for infection control purposes. An LED headlight can be worn by the operator, focusing the shadowless light in the direction of the dentist's gaze (see Figure 5.2). Care needs to be taken when working with resin composites, as these can be light-cured prematurely by intense LED outputs or the overhead operating light. An orange filter can be used to prevent this from happening when using the LED headlight, or alternatively the operating light may need to be directed away from the oral cavity when placing the resin composite restorative material.



Figure 5.3 The initial positions of the dentist and nurse during the intra-oral examination. See text for details.

5.2.3 Zoning

For infection control purposes, the dental surgery must be divided into zones for clinical notes/clean instruments and dirty instruments. It is important that these zones are designed with practicality in mind, as thoroughfare between and through them should be kept to a minimum in order to prevent cross-contamination between zones. An example in a hospital setting is shown in Figure 5.4, but variations will obviously be found in different dental practice surgeries. Dirty zones (areas that are likely to become contaminated by direct contact, aerosols, or splatter during treatment procedures) include the following:

- bracket table and handle
- dental handpiece unit, connectors, and switches
- dental chair headrest
- light handles/switch
- chair handle controls
- suction connectors
- spittoon.

These zones must be cleaned and disinfected appropriately between patients. To aid this process, most of the equipment can be covered with clear plastic wrap (cling film) or plastic sleeves, which are removed, the zones cleaned, and plastic wrap replaced between patients.

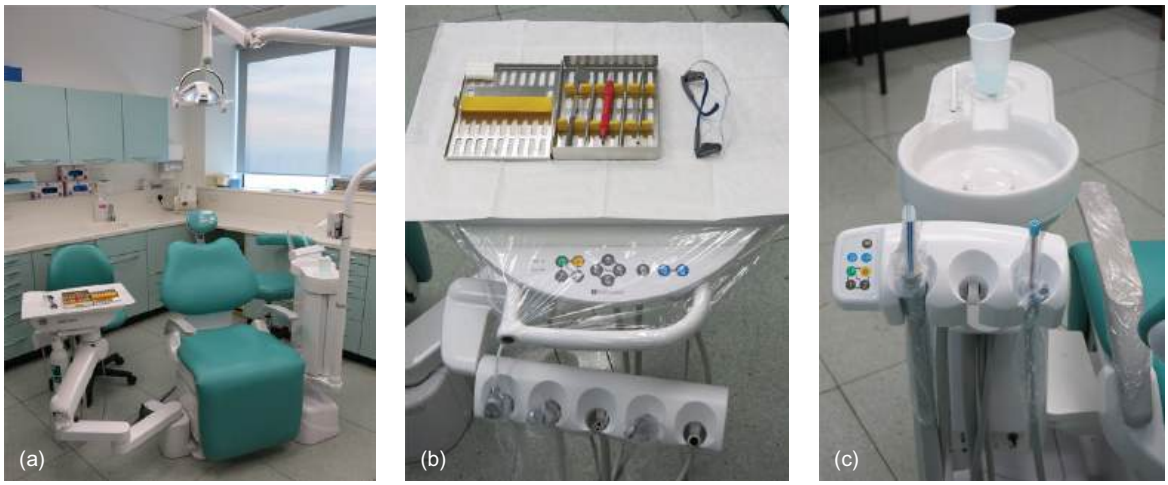


Figure 5.4 The infection control barriers placed when setting up a dental surgery. (a) An overview of the surgery. (b) The bracket table. (c) The spittoon area. Note the use of disposable suction tips and barrier protection on all tubing.

5.3 Infection control/personal protective equipment (PPE)

Current guidelines require the dentist and nurse to be dressed in short-sleeved tunic/uniforms, with no neckties that hang loosely from the neck and no jewellery (except wedding bands). Footwear should cover the tops of the feet. Standard infection control practice against blood-borne, airborne, and other body fluid-borne pathogens is outlined in Table 5.1. With regard to protecting the patient, a disposable neck towel

Table 5.1 Infection control procedures for the oral healthcare team

Infection control precautions	Comments
Hand hygiene	Soap/detergent and running water; alcohol hand gels before donning and after removing gloves
Skin dressings	Cuts, abrasions, and skin conditions protected with waterproof dressings
Sharps handling	Used needles and sharps should be disposed of by the user into rigid sharps containers Only re-sheath needles if using a re-sheathing device/single-handed technique
Personal protective equipment (PPE)	Reusable protective eyewear worn (goggles with side protection or visors over spectacles) if there is a risk of blood/body fluid splashing to the face Single-use surgical face masks provide a physical barrier to splashes to the mouth/face. They do not protect the wearer from aerosol inhalation Respirator-type masks can be used to protect against aerosol inhalation Single-use gloves: natural rubber latex (NRL, non-powdered), nitrile (acrylonitrile)/polychloroprene or Tactylon gloves should be worn (non-allergenic) NRL can result in development of latex hypersensitivity—delayed type IV (contact dermatitis, rhinitis, conjunctivitis up to 48 hours after exposure) or, less commonly, immediate type I (asthma, urticaria, laryngeal oedema, anaphylactic shock usually within 30 minutes after exposure) Single-use plastic aprons to prevent contamination of clothing
Blood/body fluid spillages	Dealt with by using hypochlorite granules and appropriate PPE
Clinical waste handling	Infectious hazardous waste disposed of in yellow clinical waste bags
Instruments	Single-use sharp instruments disposed of in appropriate sharps container. Reusable instruments must undergo decontamination prior to sterilization (see later)



Figure 5.5 Patient in a reclined position wearing eye protection and a disposable bib and neck towel.



Figure 5.6 An automatic handpiece cleaning unit which is connected to the air supply. The shield at the front is rotated out of the way and the handpiece plugged on to the connector. A detergent is then flushed through the handpiece, followed by oil, both of which come from refillable containers at the back of the unit. This is a quicker and more efficient system than using aerosols.

should be placed over them, and suitable eye protection must be worn by the patient while they are in the reclined position (see Figure 5.5).

5.3.1 Decontamination and sterilization procedures

The days of locating the sterilizing unit in the corner of the surgery are numbered, and in some countries they are long gone (see QR code image 5.1). Separate sterilization facilities for reusable equipment are required (either in a decontamination unit or a segregated area within the surgery) with contaminated entry and clean exit pathways clearly demarcated. This area should be separated into dirty, sterilization, and clean zones. Instruments need to be washed manually and placed in an ultrasonic bath to remove debris. A thermal washer–disinfector is recommended. Dental handpieces require lubrication before and after sterilization, using manual aerosols or an air-driven cleaning/lubricating unit (see Figure 5.6).

Once this decontamination cycle is completed, non-vacuum (downward/gravity displacement bench-top steam) or vacuum bench-top steam sterilization (recommended for wrapped/unwrapped, hollow-lumen items) is required before individual instruments are dried (thus preventing long-term corrosion of carbon steel instruments) and finally bagged in sterile packets and stored in the clean zone. Conventional default settings for most steam sterilizers are 134–137°C held for 3 minutes at a pressure of 2.5 bar. In hospitals, central sterilizing facilities take on this role and soiled instruments are carefully packaged.

QR Code image 5.1 Scan this QR code image with your mobile device to access UK government guidance on decontamination in dental primary care practices (HTM 01-05).

www.gov.uk/government/uploads/system/uploads/attachment_data/file/170689/HTM_01-05_2013.pdf

5.4 Patient safety and risk management

During operative dental procedures, patient safety is of paramount importance. Accidents will happen, but the risk of these occurring must be minimized or even eradicated at all costs—this is known as *risk management*. In terms of sustaining injury/harm during a dental procedure, the vulnerable areas of the patient include their eyes, airway, and soft tissues. The damage can be caused by inappropriate use/maintenance of sharp instruments, burs, and small instruments, and improper use/handling of dental materials. Table 5.2 outlines the sites, aetiology, and types of injury that might be sustained, and methods to prevent them from occurring.

5.4.1 Management of minor injuries

In all cases of injury, the patient must be informed and the events documented comprehensively in the patient's notes.

- *Eyes*: If debris enters the eyes (of the patient or a member of the dental team), immediate washing in an eye-bath with sterile water is essential, with follow-up medical care if required.
- *Airway*: If an inhalational blockage occurs, stand the patient bent at the hips and firmly slap their back to help to dislodge the object

Table 5.2 A summary of the potential sites/types of patient injury, causes, and methods used by the oral healthcare team to prevent them from occurring

Patient	Aetiology of injury	Type of injury	Prevention
Eyes	Sharp instruments slipping/burs fracturing, fragments of restoration, aerosolized dental materials/body fluids	Laceration, burn, infection	Protective glasses (with side covers) worn when patient is supine
Airway	Small instruments (fractured burs, RCT files), indirect restorations (crowns, inlays), implant components, extracted teeth/fragments	Inhalation or swallowed. Lung infection, blockage of URT airway	Rubber dam; safety chains/floss tied to small instruments; gauze squares placed intra-orally to cover back of throat
Soft tissues (lips, mucosae, tongue)	Caustic dental materials (acid etchant, hypochlorite), sharp instruments (including needles), rotary burs, heat, aggressive retraction of soft tissues, compressed air introduced into open wounds; anaesthetized tissues	Burns, lacerations, grazes, surgical emphysema	Rubber dam placed correctly with sufficient seal; proper use/maintenance of handpieces/burs; single-use needles—*do not bend*. Avoid directing air jets into broken mucosae or exposed root canals

RCT, root canal therapy; URT, upper respiratory tract.



Figure 5.7 Radiograph showing a gold crown in a patient's stomach. (Courtesy of Professor A. H. R. Rowe.)

Q5.1: Can you spot the gold crown in Figure 5.7?

from the oropharynx. Check that the object has not been sucked up the high-volume aspirator (if in use). If the object cannot be accounted for, and the patient is not sure whether they have swallowed it, a chest radiograph will be required to help to localize the foreign object in the bronchi, lungs, oesophagus, or stomach (see Figure 5.7). Follow-up medical attention will be required, especially for the former, as a bronchoscopy may be indicated.

- *Soft tissues*: Haemostasis of cuts and abrasions must be achieved. Cuts to the lips or mucosae may need sutures. Caustic sodium hypochlorite burns (caused by a leaking rubber dam in root canal therapy) may require medical attention and a suitable dressing to promote healing without scarring. Patients with inferior alveolar nerve anaesthesia should be warned about the persistent numbness that will occur for several hours after treatment, and advised to take care to avoid biting/chewing their lower lip or consuming hot food or drink in case they burn themselves.

5.5 Dental aesthetics and shade selection

The advent in the early 1960s and continued development of direct adhesive, tooth-coloured restorative materials have raised patients' expectations with regard to high-quality aesthetic results when it comes to restoring teeth to form and function, especially in the anterior 'aesthetic zone' (UR4–UL4). Indeed, we are now at a stage in minimally invasive conservative dentistry where all operative interventions should be aesthetic in nature. An aesthetically pleasing dental result depends on numerous complex and interlinked factors, including:

- the three-dimensional shape and position of the tooth
- the surface form and finish of the tooth
- the inherent shade of the tooth
- the morphology of the surrounding periodontal and facial tissues
- the occlusal plane and relationship with adjacent/opposing teeth.

The dental materials that can offer shades and characteristics that closely match natural tooth aesthetics are, in descending order:

- dental porcelains (indirect—made by a dental technician in a laboratory from casts poured from dental impressions and finally cemented in/on the tooth)
- dental resin composites (direct—placed at chairside; indirect—made in the laboratory)
- glass ionomer cements (direct—including conventional, resin-modified, and polyacid-modified composites).

5.5.1 Colour perception

The human brain is able to perceive intrinsic physical properties of incident light sensed by the eye. In order to communicate these, colour scales have been devised. One of the earliest was the system from A. H. Munsell, which consisted of three elements—the dominant wavelength (*hue*), the excitation purity (*chroma*), and its luminous reflectance (*value*) (see Table 5.3).

However, the human tooth is heterogeneous, and the pulp, dentine, enamel, their interfaces, and relative thicknesses along with their changing histological structure all play a part in overall perception of the shade by affecting the interaction between incident light and tooth structure. When attempting to mimic the shade of a tooth, the inherent structure of the restorative material and how it affects the physical characteristics of incident light must be considered (i.e. the reflectance, translucency, opacity, and fluorescence). As the Munsell classification does not take these factors into account, another classification was devised by the Commission Internationale de l'Eclairage (CIE)—the CIE L* a* b* colour space. This classification applies the amount of red, green, and blue colours used. L* measures lightness (0–100, black–white) and CIE a* and b* the relative hue and chroma (a*, levels of red(+)-green(–); b*, levels of yellow(+)-blue(–)). This system permits quantification and calculation of colour numerically, expressed in units that can be related to visual perception and clinical significance. This scale is used in research studies into colour and also the electronic shade guides (see later). The properties of the restorative material that can help to mimic these interactions include the following:

- *Refractive index*: Refraction is the change in direction of light due to it entering a medium of different density. Different materials have

Table 5.3 The intrinsic physical properties of light perceivable by the human eye according to the Munsell colour space classification

Property of light	Comments
Hue (wavelength)	Wavelength of coloured light translated as its actual colour (e.g. red, green, blue, etc.). Figure 5.8 shows the Vitapan Classical shade guide tabs arranged in the four-lettered groups according to hue: A, more reddish-brown; B, yellow; C, yellow-grey; D, more reddish-grey
Chroma (saturation)	Strength/dominance of the colour (hue). Light or dark shade. Figure 5.8 shows the Classical shade tabs arranged with increasing chroma (1–4) in each of the hue (colour) groups (A–D)
Value (greyness)	Luminosity of the colour (the level of black or white—its *greyness*). Figures 5.9 and 5.10 show the shade tabs arranged in descending order of value. Enamel thickness is the major contributor to the value of a tooth



Figure 5.8 The Vitapan Classical shade guide tabs, arranged in four groups, A–D, according to hue (colour) and, within each hue, a range of increasing chroma, 1–4.



Figure 5.9 The Vitapan Classical shade guide tabs arranged in order of value, from high (whiter) to low (darker).

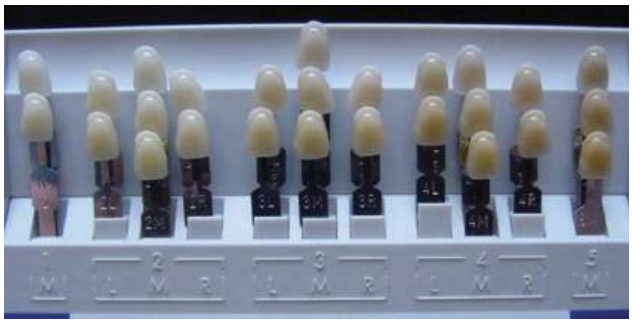


Figure 5.10 The Vitapan 3D-Master shade guide with value graded from 1 to 5 (light to dark), three chroma choices within the value groups 2, 3, and 4, and the individual hue (yellow-red).

different refractive indices. The difference between the refractive indices of two materials determines how opaque or translucent one appears when observed from the other. The opacity of a resin composite depends on the difference between the refractive index of the resin composite material and the refractive index of air (refractive index of vacuum/air, 1.0; water, 1.33; enamel, 1.63; dentine, 1.54; cementum, 1.58).

- *Scattering coefficient*: Scattering is the loss of light due to the reversal of its direction. The scattering coefficient varies with the wavelength of the light and the nature of the colourant layer of a composite. A composite with a higher refractive index (dependent on the polymer matrix, size/type/amount of filler particles, and aluminium/zirconium/titanium oxide opacifiers present) will appear more opaque.
- *Absorption coefficient*: Absorption is the taking up of light by a material. The absorption coefficient also depends on the wavelength of light and the nature of the colourant layer of the resin composite. The higher the absorption coefficient, the more opaque and intensely coloured the resin composite will appear.

When restoring a tooth with a direct resin composite, it is advisable to use incremental layering techniques in order to copy these variations that occur within the tooth. In a young tooth, the thicker enamel will affect the value and hue, with the dentine contributing to the chromatic aspect of its shade. As the tooth ages and the enamel thins, the dentine and pulp will have more of an influence on the hue, with enamel contributing primarily to the value of the final shade. The enamel–dentine junction (EDJ) contributes to the fluorescent characteristics (which can be added in a layering technique using coloured tints and stains). The quality of the surface finish and form will affect the reflectivity of incident light; using a flowable, translucent low filler particle content resin composite will assist in producing a smooth, blemish-free surface after careful polishing (see later). Some manufacturers provide their own shade guide with their layering resin composite system to simplify the choice of dentine and enamel shades, often calibrated to the Vitapan Classical system.

5.5.2 Clinical tips for shade selection

- Avoid brightly coloured neck towels. It is helpful to use a light blue neck cloth. Select the shade before placing rubber dam, as teeth become lighter as they dry out.
- Ask the patient to remove bright lipstick or other obstacles, such as hoods or hats, which may affect incident light/shadows.
- Schedule aesthetic restoration appointments early in the morning if at all possible, to avoid eye fatigue.
- The light source should be diffuse, not direct. Use natural daylight where possible. Try to avoid conventional fluorescent light sources.
- Fan the shade guide past the patient's mouth and pick the closest tab. Do not stare. Rest your eyes occasionally. After you have selected the hue, squinting/half-closing your eyes will aid the determination of the value.
- Cold sterilize shade guides to prevent damaging them.
- Shade taking should be carried out after prophylaxis but before tooth isolation and preparation, since dehydration can lighten the natural tooth shade as light is reflected from pore spaces within the enamel as it dries out.
- Select the body shade by examining the centre portion of the tooth (see Figure 5.11). Check with the mouth open and closed (lips parted) for anterior dentition.
- Choose the resin composite shade that most closely approximates the centre portion of the shade guide (see Figure 5.11). Custom-made resin composite shade tabs can be constructed from the specific brand of material used by the dentist. Using glass slabs, various thicknesses of each shade of resin composite can be photo-cured and labelled, and used subsequently for clinical shade selection.
- Place directly and cure the chosen shade of resin composite on the tooth surface to be restored, using the correct thickness of material (see Figure 5.11). No etch or bond is required. The patient can use a face mirror to help them to check and agree to the shade of material selected.

As well as the visual shade guides depicted in Figures 5.8, 5.9, and 5.10, there are electronic shade detectors which shine an incident beam of light on to the relevant tooth and then numerically analyse the reflected light. Care has to be taken when assessing particularly translucent incisal edges, as these instruments can misreport the translucency as a greyer overall shade as a significant proportion of the light passes through the tooth rather than being reflected back.

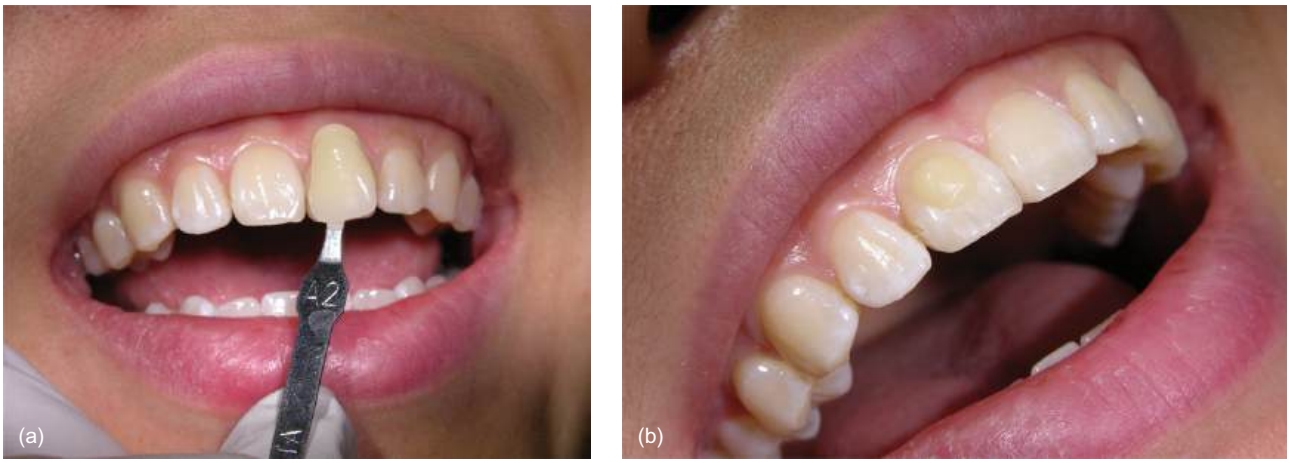


Figure 5.11 Shade assessment of UR1 using a shade tab (left image) to assess the shade from the body of the tab and tooth, and then placing the resin composite on the unetched labial surface of the tooth to make the final assessment, ensuring that the tooth is not dehydrated.

5.6 Moisture control

This is the ability to regulate the fluid environment within the oral cavity and around the individual teeth that is being operated on. Fluids include water, saliva, gingival exudate, and blood.

5.6.1 Why?

Moisture control is required:

- to permit proper placement of all restorations—excessive fluids have a detrimental effect on the adhesion and/or physical properties of all direct plastic restorations, but especially those that rely on adhesive bonding
- to prevent contamination of the restorative procedure—caries removal, direct pulp capping, root canal therapy, and adhesive bonding
- to improve patient comfort—removing accumulated fluids produced by dental handpieces, 3-1 syringes, and salivary flow.

5.6.2 Techniques

- *Aspiration*: Fluids can be evacuated with a single-use saliva ejector by the nurse, the dentist, or even the patient (see Figure 5.12). Plastic tips can be bent into a curved shape to adapt the contours of the lips and oral cavity, thereby accessing the buccal or lingual sulci while simultaneously retracting these tissues.
- *Cotton wool rolls (cellulose pads)*: Used to absorb fluids and retract the lips or cheeks. Placed over salivary duct orifices, upper buccal sulci (parotid ducts), and lingual sulci (submandibular and sublingual ducts; see Figures 5.12 and 5.13). Cotton wool rolls should be rotated into the buccal, labial, or lingual sulci to aid their retention. If still dry at the end of treatment, moisten them with a few drops of water from the 3-1 syringe to facilitate their removal and avoid them sticking to the mucosa.



Figure 5.12 Series of images showing how to successfully place a cotton wool roll into the maxillary right buccal sulcus. Note the way that it is rotated into position, thereby ensuring that it will stay in place. A blue flanged disposable saliva ejector tip has been placed lingually to help to retract the tongue.



Figure 5.13 A cellulose pad has been used to absorb moisture from the right parotid duct within the maxillary right buccal sulcus.

- *Rubber dam*: A thin latex (or equivalent non-latex—especially important for patients who may be sensitized to latex) perforated sheet that isolates single teeth or groups of teeth, so allowing the best control of moisture. Its advantages and disadvantages are listed in Table 5.4, and the equipment required to place rubber dam isolation is described in Figures 5.14, 5.15, 5.16, and 5.17.

5.6.3 Rubber dam placement: the practical steps

Figures 5.16 and 5.17 show how a rubber dam can be placed on and then removed from an upper molar using either a wingless (Figure 5.16) or winged clamp (Figure 5.17). Some patients may need local anaesthesia/topical anaesthesia to dull the pain of the clamp on the tooth/periodontium. If a matrix band is required to help to contour the final restoration, the rubber dam clamp may need to be removed as it will get in the way of the circumferential or sectional matrix system (see later). In this situation, the matrix band will take over the role of the clamp in holding the dam in position while the material is packed into place.

For anterior teeth, there is less need to use clamps, as tight contact points can be sufficient to retain the well-adapted dam interdentally, or else a thin strip of rubber dam cut from the peripheral excess/proprietary rubber wedgets can be used by 'flossing' them beyond the contact points. Floss ligatures may also be used. If the contact points are too tight initially for even floss to pass, interdental pre-wedging with a wooden wedge for 5 minutes before placing the dam will usually cause enough tooth movement within its periodontal ligament (and that of the adjacent tooth) to allow the dam to pass between. Careful use of the anterior Ferrier clamp is required so as not to traumatize the gingival margins.

Table 5.4 The advantages and disadvantages of placing and using rubber dam for moisture control

Advantages	Disadvantages
Controlled isolation of the teeth from fluids (reducing adverse effects on bonding/physical properties)	More difficult to communicate with the patient
Reduced bacterial contamination (caries removal, pulp capping, RCT)	Some patients dislike the feeling of claustrophobia (may be reduced by relieving the dam away from the nasal passages)
Patient airway protection from inhalation/swallowing instruments (restorations, burs, endodontic files, wedges); orofacial soft tissue protection from chemical agents (acid-etch gel, dilute hypochlorite solution used in RCT)	Attaching the rubber dam to teeth with clamps can cause pain and some post-operative pain for several hours
Can act as barrier protection from fluid-borne pathogen transfer from patient to dentist	Poorly fitted clamps can cause damage to ceramic crowns (chipping)
Positive psychological experience/improved comfort for some patients. A feeling of detachment/separation from the operation	Care is needed to check medical history for latex allergies for both patient and dentist. Non-latex containing dams are available
Can aid in tissue retraction and so improve direct vision—keeps oral mucosae and tongue separate from the operation field	In inexperienced hands, rubber dam placement can be difficult and time-consuming. However, once it has been mastered, this problem is alleviated
Once in place, rubber dam can speed up the procedure for the dental team	

RCT, root canal therapy.

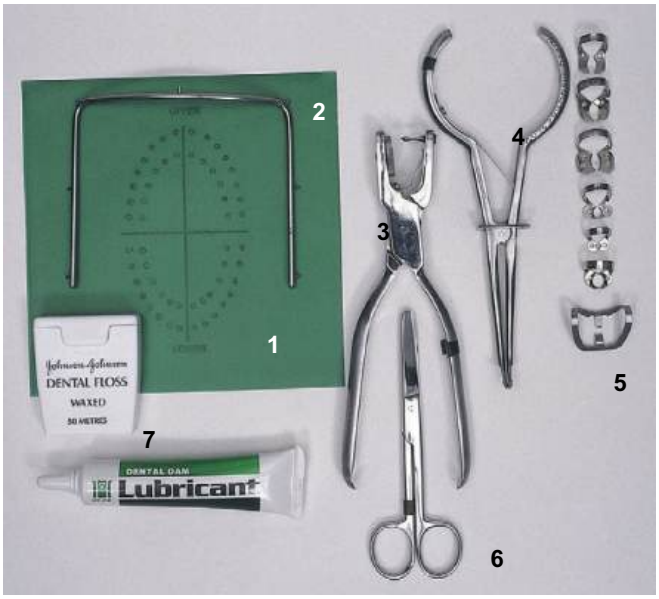


Figure 5.14 Rubber dam equipment comprising:

1. A 15-cm-square dark green rubber sheet (with the position for the perforation holes stamped on using an ink stamp). The rubber resists tearing and grips the tooth surface. Dark colours are used to contrast with the teeth, and torn fragments, often interproximally, can be detected and removed.
2. Metal rubber dam frame holds the free edges away from the face/mouth. The dam is stretched around the tines on the frame itself, keeping it under tension.
3. Rubber dam hole punch, which creates a clean cut perforation of three diameters (corresponding to incisors, premolars, and molars).
4. Rubber dam clamp forceps, used to place, adjust, and remove clamps from teeth.
5. Rubber dam clamps—metal clips that grasp the coronal neck of the tooth holding the dam down. Can be winged or wingless (see Figure 5.15).
6. Scissors to cut excess dam away from the nose and, when removing the dam, interproximally.
7. Waxed dental floss and a water-based dam lubricant—both aid the transition of the dam through tight contact points. Floss can act as a ligature, tying the dam down at the gingival margin.

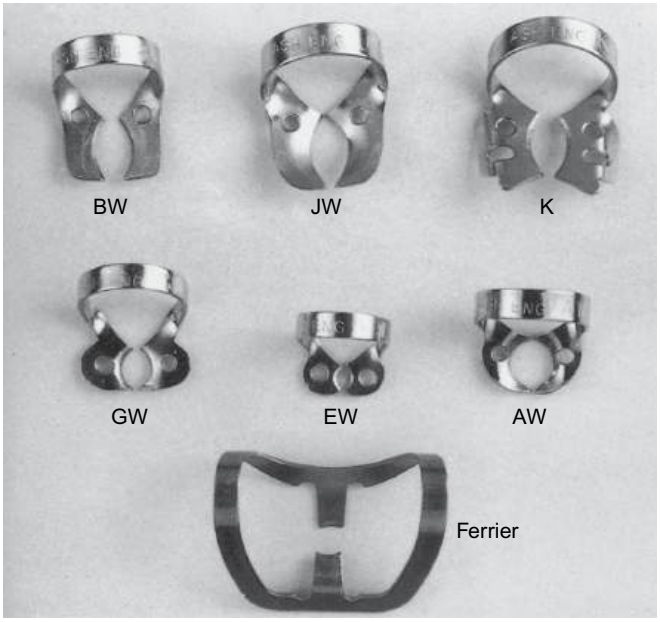


Figure 5.15 A close-up view of metal rubber dam clamps, consisting of two jaws joined by a curved connecting arm. The two holes accommodate the clamp forceps, which allow the opposing jaws to be separated when placing/removing them over the crown bulbosity. BW, JW, K, and AW clamps are molar patterned (AW is more retentive for partially erupted crowns), GW is used for canines/premolars, and EW is configured for any other smaller tooth. The K clamp is winged—it has two extensions on to which the dam can be placed prior to inserting the dam and clamp together on to the tooth (useful when isolating just one tooth in the arch). The remaining wingless clamps must be first placed on to the teeth, followed by the manual application of the dam over them. The Ferrier clamp is used for retracting gingivae when isolating anterior teeth, working in the cervical area (care is needed not to traumatize the gingivae and cause blood contamination of the field; see Chapter 7).



Figure 5.16 **(a)** A wingless JW clamp placed on the UL6. Clamp forceps were used to open the jaws of the clamp sufficiently to allow it to pass over the bulbosity of the crown. Note the floss tied through both clamp holes and coiled around the connector arm. This makes any clamp fragments retrievable if the connector arm fractures under tension when in use. The connector arm is placed distally to prevent the clamp from blocking access to the tooth.



Figure 5.16 **(b)** The dam is stretched gently by the dentist to widen the lubricated perforation, and the free end of the floss is passed through the hole.



Figure 5.16 **(c)** The dam is stretched over the connector arm first and then over the two clamps embracing the tooth. The nurse can help by keeping the floss taut throughout the procedure.



Figure 5.16 **(d)** The dam is then 'knifed' though the contact areas of the remaining teeth to be isolated in the UL quadrant.



Figure 5.16 **(e)** Floss is used to work the dam down through the mesial contact area of the UL6.



Figure 5.16 **(f)** A strip of dam has been wedged in the distal contact of the UR3 to hold the anterior portion of the dam in place. Sometimes a naturally tight contact may suffice. The dam has yet to be inverted around the cervical margins of the teeth.



Figure 5.16 **(g)** Alternatively, a floss ligature could be used to the same effect, using a flat plastic instrument to ensure palatal placement apical to the maximum bulbosity of the crown and inverting the dam simultaneously. It will not always be clinically necessary to isolate so many teeth if only working on the UL quadrant.



Figure 5.16 **(h)** When removing the dam, it is pulled buccally and cut interdentally using scissors. Note the operator's finger beneath the dam protecting the patient's lip, and the paper towel placed between the dam and the patient's face for comfort (make sure that the nasal airway is not covered by the dam or paper towel). The frame, dam, and clamp are removed together and checked for completeness.



Figure 5.17 **(b)** Clamp and rubber dam being secured to the UL6 as one, using the clamp forceps (note the two pairs of holes to engage the forceps). The nurse can assist the process by gently retracting the loose rubber dam edges to aid vision. Again, the connector arm is placed distal to the tooth being operated upon so as not to block access to instrumentation/visibility.



Figure 5.17 **(a)** K clamp (winged molar) with the wings engaged in the lubricated perforation in the dam, outside the mouth.



Figure 5.17 **(c)** The dam is disengaged from the wings using a flat plastic instrument, and the contact areas will be flossed to ensure interdental adaptation of the dam.

Q5.2: Can you spot an important clinical omission in Figure 5.17?

5.7 Magnification

Visual magnification aiding clinical examination/operating can be achieved using *magnifiers* or *dental loupes*. A selection of these are shown in Figure 5.18. In terms of optical physics, there are two aspects to consider with regard to their clinical application—the *depth of field (or focus)* and the *field of view (or field width)*.

- *Depth of field (DoF)*: This equates to the distance the operator can move towards or away from the object tooth while keeping the image in clear focus. If this range is limited, procedures become tiring as the operator and patient must always remain at a fixed distance from each other with minimal leeway of movement of either person before the view becomes blurred. For a given magnification, the DoF depends on the f-number of the lens (aperture diameter); ↑ f-number (↓ aperture) leads to an ↑ DoF.
- *Field of view (FoV)*: The angular extent of the observable image viewed through fixed-magnification loupes. Normal unaided human binocular vision (with depth perception) permits 140° FoV. With increasing magnification, this is greatly reduced—for example, 3× magnification with loupes may only permit a clear, sharp view of the object tooth and one either side (depending on the precise optics of the loupes). This can make the introduction of dental instruments into the operating field difficult, and will require the dentist to look away from the loupes' view to do this.

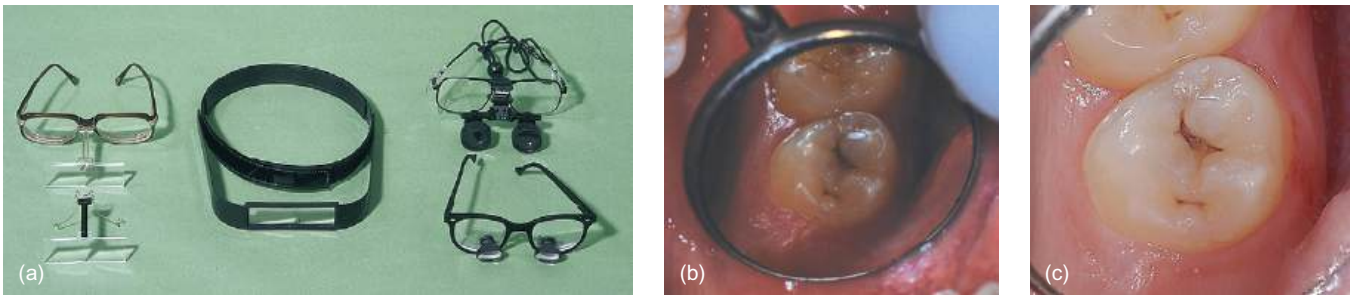


Figure 5.18 **(a)** A selection of magnification loupes. The first three are the cheapest and have the simplest optics, clipping on to existing spectacles/safety glasses or being secured with a head loop. The two on the right are more expensive. The top right pair have interchangeable lens blocks, and due to their weight require a head cord for retention. Those on the bottom right have the dentist's fixed interpupillary distance built in, as the lenses are cemented through the prescription/safety spectacle lenses. As the magnification optics are closer to the eye, this results in a greater field width and depth of field, but at increased expense. **(b)** The clinical view of an inadequately lit maxillary premolar examined without any magnification. **(c)** The same tooth now lit effectively using an LED head-mounted light along with 2× magnification. More detail can be observed regarding the state of the mesio-distal fissure. ((b) and (c) Courtesy of L. Mackenzie.)

For general dentistry, 2–3× magnification is usually satisfactory, but for endodontics, 4–5× magnification can be used to help to detect root canal orifices and fine root canals. Good visual discrimination in dental procedures requires good depth perception, which in turn requires binocular function. To get the most benefit from the increased magnification, good-quality illumination is also vital. Modern magnification systems often can be fitted with LED lights that are integrated on the frame above the lenses and can be focused on the same focal plane as the lenses. Orange light filters should be used to prevent premature light curing of resin-based materials.

It is the authors' opinion that using magnification to aid the execution of most intra-oral dental procedures is an essential prerequisite for practising high-quality minimally invasive operative dentistry. It is important that the skill of using loupes is learned early on in the clinical skills laboratory, and then transferred to the clinical situation. The physical adjustments made by the operator when using loupes involve practice with a learning curve.

5.8 Instruments used in operative dentistry

Instruments are used to examine, clean, cut, and restore teeth. The main types of cutting instruments are either hand-held or rotary instruments driven in a handpiece. Other equipment includes fibre-optic lights for illumination, light-curing systems used for polymerization of resin-based materials, new instruments for tooth cutting/caries removal, and ultrasonic scalers (see Table 5.5). These instruments may be reused

Table 5.5 Tooth-cutting/caries removal technologies, the substrates acted on, and the mechanism of action

Mechanism	Substrate affected	Tooth-cutting technology
Mechanical, rotary	Sound or carious enamel and dentine	SS, CS, diamond, TC, ceramic, and plastic burs*
Mechanical, non-rotary	Sound or carious enamel and dentine	Hand instruments (excavators, chisels), air abrasion (alumina/bioactive glass), air polishing (sodium bicarbonate/bioactive glass /beta-TCP)†, ultrasonics, sono-abrasion
Chemo-mechanical	Carious dentine	Caridex (of historical value, not available now for clinical use), Carisolv™ gel (amino acid based), Papacarie® gel (papain based), experimental pepsin-based solutions/gels
Photo-ablation	Sound or carious enamel and dentine	Lasers
Others	Bacteria	Photoactive disinfection (PAD), ozone

*Works only on carious dentine.

†Primarily used for stain removal.

SS, stainless steel; CS, carbon steel; TC, tungsten carbide.

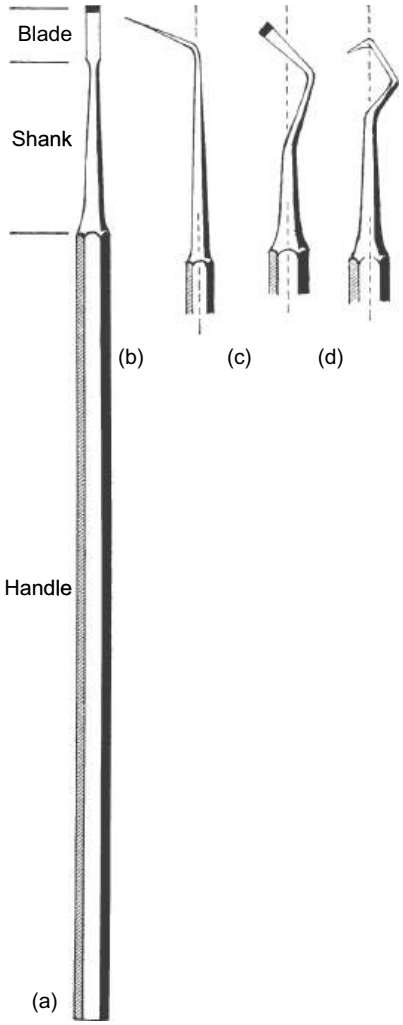


Figure 5.19 **(a)** The parts of a hand instrument. **(b)** A straight dental probe with a single bend taking the working tip well away from the handle's long axis, thus aiding direct vision. **(c)** Two bends in a hatchet. **(d)** Three bends in a Briault probe. Medical-grade stainless steel cutting edges can be resharpened using a sharpening stone with oil lubricant (fine edge) or a dry abrasive disc (coarse edge). Tungsten carbide tips should be sent to the manufacturer for resharpening.

after suitable decontamination and sterilization procedures, or else are disposable, single-use items.

5.8.1 Hand instruments

Manufactured from medical-grade stainless or carbon steel (sometimes with tungsten carbide brazed to the cutting edges for increased longevity of sharpness), the majority of hand instruments are designed with a handle, shank, and blade configuration (see Figure 5.19).

Hand instruments can be used for the following purposes:

- Oral examination (mouth mirror, selection of dental probes, pair of tweezers):
  - *Mouth mirrors*: front- or rear-surfaced (see Figure 5.20)

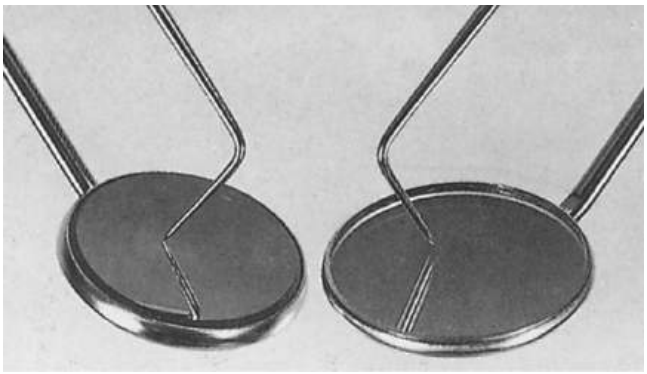


Figure 5.20 Front- and rear-surface mouth mirrors. The left mirror (front-surface) gives a clearer image than the rear-surface 'double' image (right), but is more prone to scratch damage during use and sterilization procedures.

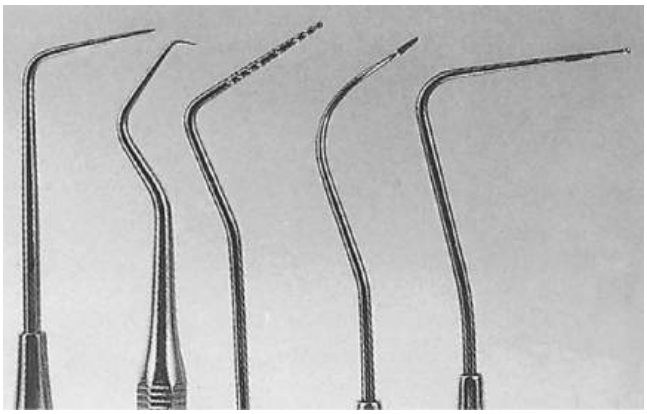


Figure 5.21 A selection of dental probes (from left to right): straight, Briault, Williams, Nabers, and CPITN probes.

Q5.3: In which clinical situations might you use each of the probes shown in Figure 5.21?

  - *Dental explorers/probes*: sharp probes (straight and Briault) are used for checking restoration margins and carious dentine to be excavated within a cavity. Round-ended/blunt periodontal probes are used for periodontal tissue examination and assessing the roughness of enamel/occlusal/smooth surfaces during caries lesion detection/examination (see Figure 5.21)
  - *Locking tweezers*: to place/remove cotton wool rolls, and remove larger pieces of intra-oral debris.
- With the advent of modern infection control guidelines, disposable examination instruments are now manufactured from plastic to allow single use only (see Figure 5.22).
- Periodontal scaling: a selection of *hand-held scalers* (see Figure 5.23) to remove supra- and subgingival calculus deposits.
- Caries removal (excavators, chisels/hatchets/hoes):



Figure 5.22 Disposable, single-use, plastic examination instruments—mirror, probe, and tweezers—in their sterile packaging.

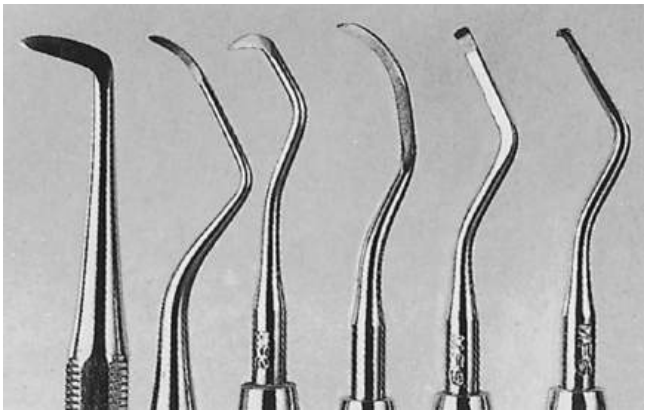


Figure 5.23 A selection of hand-held periodontal scalers (blade and shank visible). Note the angulation of the heads and the extent to which the cutting blades are offset from the long axis of the handle.

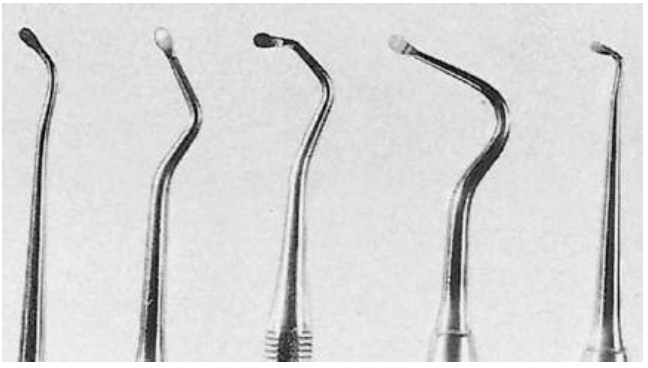


Figure 5.24 A selection of hand excavators (blade and shank visible). Note the different ovoid blade sizes and offset angulations.

Q5.4: What are the advantages of having angulations between the working head and shank of the instruments pictured in Figure 5.24?

- *Excavators*: instruments with a discoid/ovoid blade sharpened to a cutting edge are used to remove caries and soft temporary restorations. Can also be used to shape plastic restorative materials (see Figure 5.24)
- *Chisels, hatchets and hoes*: used to remove unsupported enamel/bevel cavity margins (especially where access for rotary instruments is limited). Hatchets and hoes are similar to chisels in having a straight bevelled cutting edge, and are always angled or contra-angled. They differ from one another in that the cutting edge of the hatchet is in the plane of the shank (like an axe), whereas the cutting edge of the hoe lies in an axis at right angles to this plane (as in a gardener's hoe; see Figure 5.25).

- Handling restorative materials (flat plastics, condensers (pluggers), carvers):
  - *Flat plastic instruments*: used for conveying, placing, and shaping plastic materials that do not require heavy pressure. Usually made of stainless steel, but for composite placement, Teflon-coated or titanium nitride-coated blades confer a useful non-stick property (see Figure 5.26)
  - *Condensers (pluggers)*: smooth surface instruments for packing plastic restorative materials into cavities under pressure (eliminating voids) (see Figure 5.26)
  - *Carving instruments*: sharp/semi-sharp blades that carve the shape/contour of the final restoration by a cutting/scraping action. Different patterns exist (e.g. Ward's, Half-Hollenback) (see Figure 5.26).

5.8.2 Rotary instruments

Dental burs, stones, and cutting and polishing discs are small instruments ('drill bits') gripped firmly by a chuck in a handpiece ('dental drill') powered directly by compressed air, the *air turbine*, or by a separate motor that is either air or electrically driven, the *low-speed* handpieces.

- *Air-turbine ('high-speed') handpieces*: clockwise rotary speeds between 250 000 and 500 000 revolutions per minute (rpm) but relatively low torque, achieved by a small air-driven turbine or rotor mounted in bearings in the head of a contra-angled handpiece. Burs

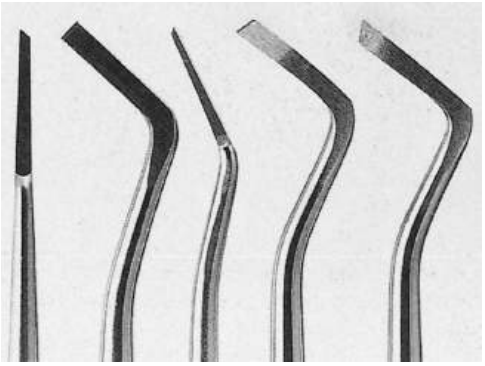


Figure 5.25 A selection of straight and angled chisels (from left to right): straight, hatchet, hoe, and a pair of double-ended gingival margin trimmers.

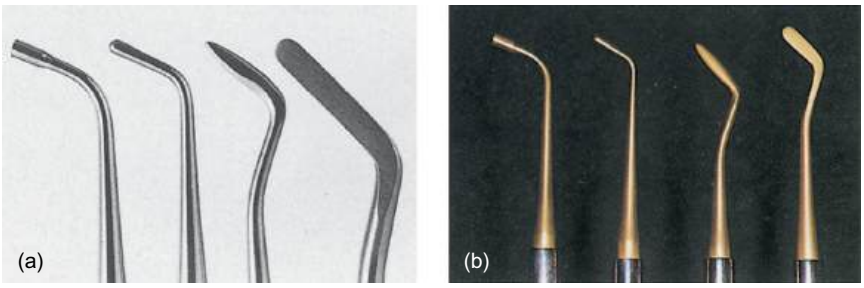


Figure 5.26 Stainless steel and titanium nitride-coated (non-stick, gold-coloured) instruments for conveying, placing, and shaping composite materials. Left to right in both images: Guy's pattern condenser (plugger), burnisher (to adapt material to cavity margins), carving instrument (Half-Hollenback pattern), and flat plastic.



Figure 5.27 Air-turbine handpiece in operation showing the directed water spray over the tip of the bur and the fibre-optic light.



Figure 5.28 A selection of low-speed handpieces. From the left:

1:1 contra-angled handpiece used for most procedures. Latch-grip burs are used. Commonly identified with a blue-coloured band on the shank of the handpiece and a blue dot on the head. Speed in the range 400–40 000 rpm.

1:4 speed-increasing handpiece. Friction-grip burs. Operates at 16 000–160 000 rpm. Commonly identified with a red band. Useful for finishing cavity preparations and also finishing restorations.

7:1 speed-reducing handpiece. Latch-grip burs. Used for drilling pin holes and other procedures where slow speed is indicated. Operates at 550–5500 rpm and is commonly identified with a green band.

Straight handpiece that takes straight burs (may be modified to take latch-grip burs). A 1:1 handpiece is identified with a blue band; speed-reducing is identified with a green band. Used to trim temporary restorations and other similar procedures. Usually used outside the mouth.

A fibre-optic light system built into the head of a contra-angled low-speed handpiece.

are held via a friction-grip shank, the tip shrouded in water spray and illuminated with a fibre-optic light (optional; see Figures 5.27 and 5.28).

- *Low-speed handpieces*: two-piece system comprising an electric or air-driven motor coupled to a contra-angled or straight handpiece (rotating clockwise or anticlockwise), with water spray and fibre-optic light; it has lower speed but higher torque than the air-turbine handpiece (see Figure 5.28).
- *Dental burs* (see Figures 5.29 and 5.33), *stones* (see Figure 5.30), and *finishing burs/discs* (see Figures 5.31 and 5.32): gripped in handpieces by a quick-release clamping chuck (friction-grip diamond grit/tungsten carbide (TC) cutting blades for air-turbine handpieces; latch-grip carbon steel/diamond grit/TC/plastic cutting blades for low-speed handpieces).

5.8.3 Using hand/rotary instruments: clinical tips

- Rotary instruments 'cut' hard dental tissues by chipping and smashing mechanically at dental hard tissue ultrastructure. Therefore the resulting prepared surface is often damaged at the microscopic level (see Figure 5.34). This effect has implications when considering the suitability of these surfaces for adhesive bonding (see later).
- Rotary instruments generate heat at the cutting interface (dissipated by the water spray from both high- and low-speed handpieces), as well as vibration, pressure, and noise (all of which are unpleasant for patients). They provide a reduced tactile feel compared with hand instruments, and therefore cutting is guided visually. As a general

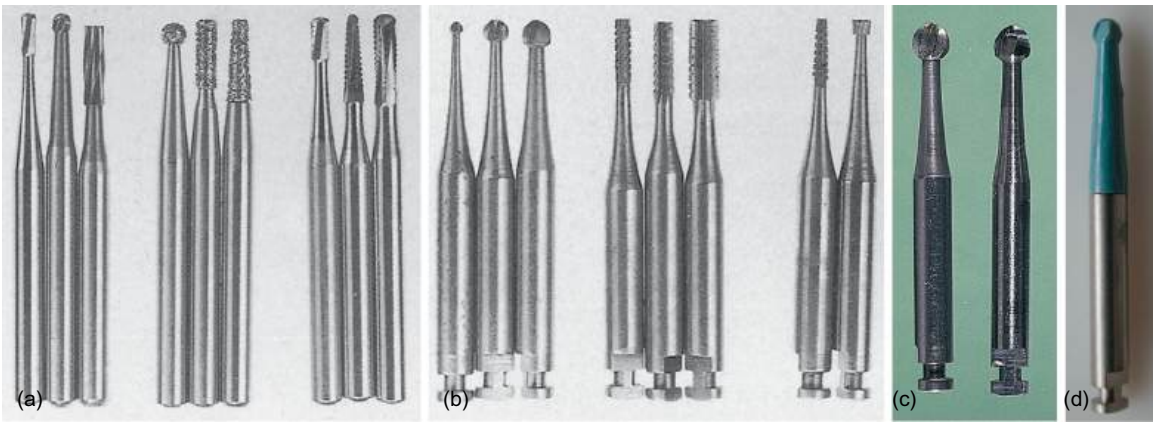


Figure 5.29 A series of eight sets of burs (first three sets, friction-grip for the air turbine; next five sets, latch-grip for the low-speed handpiece).

(a) Friction-grip: tungsten carbide (TC) ×3, diamond grit (round, straight, tapered) and metal-cutting (serrated cross-cut TC blade) burs. TC and diamond used for cutting sound enamel and dentine, removing existing restorations. Diamond burs can be used to cut ceramic.

(b,c) Latch-grip, carbon steel: three sizes round (rose-head), three sizes straight cross-cut fissure, tapered cross-cut fissure, and inverted cone burs (now rarely used clinically). Burs numbered according to size/diameter of cutting head. Carbon steel heads used for cutting carious dentine but blunt quickly, corrode if not dried after sterilization, now often used as a single-use bur.

Similar size carbon steel and TC rose-head burs compared. The TC head is a slightly different shape and is attached to the steel shank. Greater longevity and autoclavable.

Developments have included the latch-grip, rose-head pattern PKK (polyketone-ketone) plastic bur **(d)**, for 'self-limiting' carious dentine removal, single-use.

Q5.5: What might be the advantages and disadvantages of single-use, disposable burs?

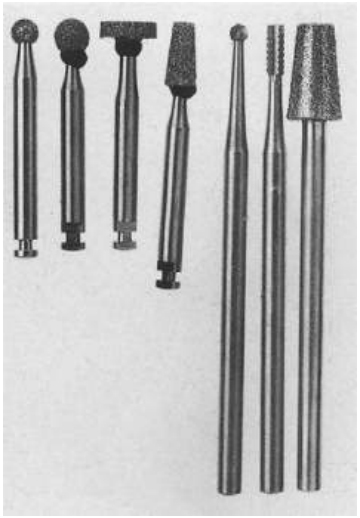


Figure 5.30 A selection of four dental stones mounted on latch-grip short shanks (abrasive carborundum green stones) and three long-shank instruments. Far right is a diamond-coated cone-shaped bur for use in the dental laboratory or rarely at the chairside, for coarse adjustment of indirect restorations and appliances, used at medium speed in a straight handpiece.



Figure 5.31 Finishing burs, stones, and points for dental amalgam. From the left: five plain-cut latch-grip steel finishing burs, two mounted white (alundum) stones, three mounted abrasive rubber points from coarse to fine, and a mounted abrasive rubber cup.

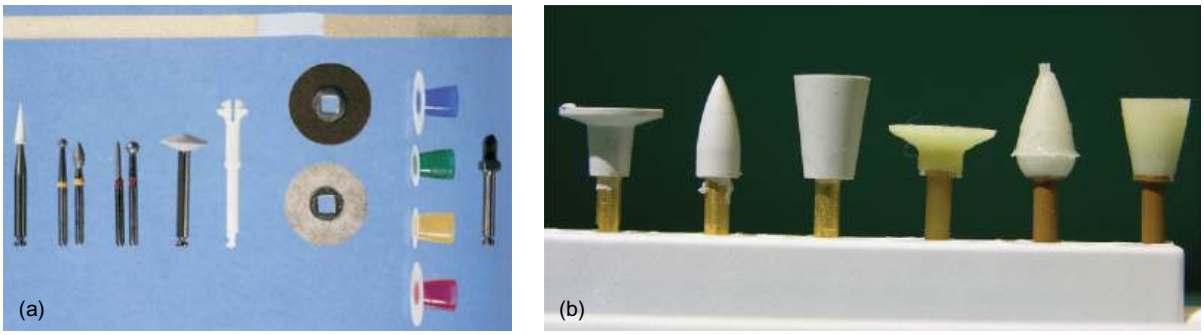


Figure 5.32 **(a, b).** Resin composite finishing burs/discs. Across the top is a proximal plastic finishing strip with a blank area in the middle to facilitate passage through the contact point between the teeth. On the right portion the abrasive is coarse and on the left it is fine. From the left: a mounted fine white stone, two medium-grit composite finishing diamonds (yellow band), two fine-grit composite finishing diamonds (red band), a mounted abrasive rubber disc, a mandrel for the two abrasive single-sided flexible discs, which are snap-fitted on to the mandrel, and four colour-coded flexible abrasive discs (coarse to fine impregnated grit/diamond) mounted on plastic stubs which fit the mandrel to the right of the picture. In the right image is a series of rubber/resin discs, points, and cups impregnated with silica grit, enabling the finishing of resin composite and glass ionomer cement restorations.

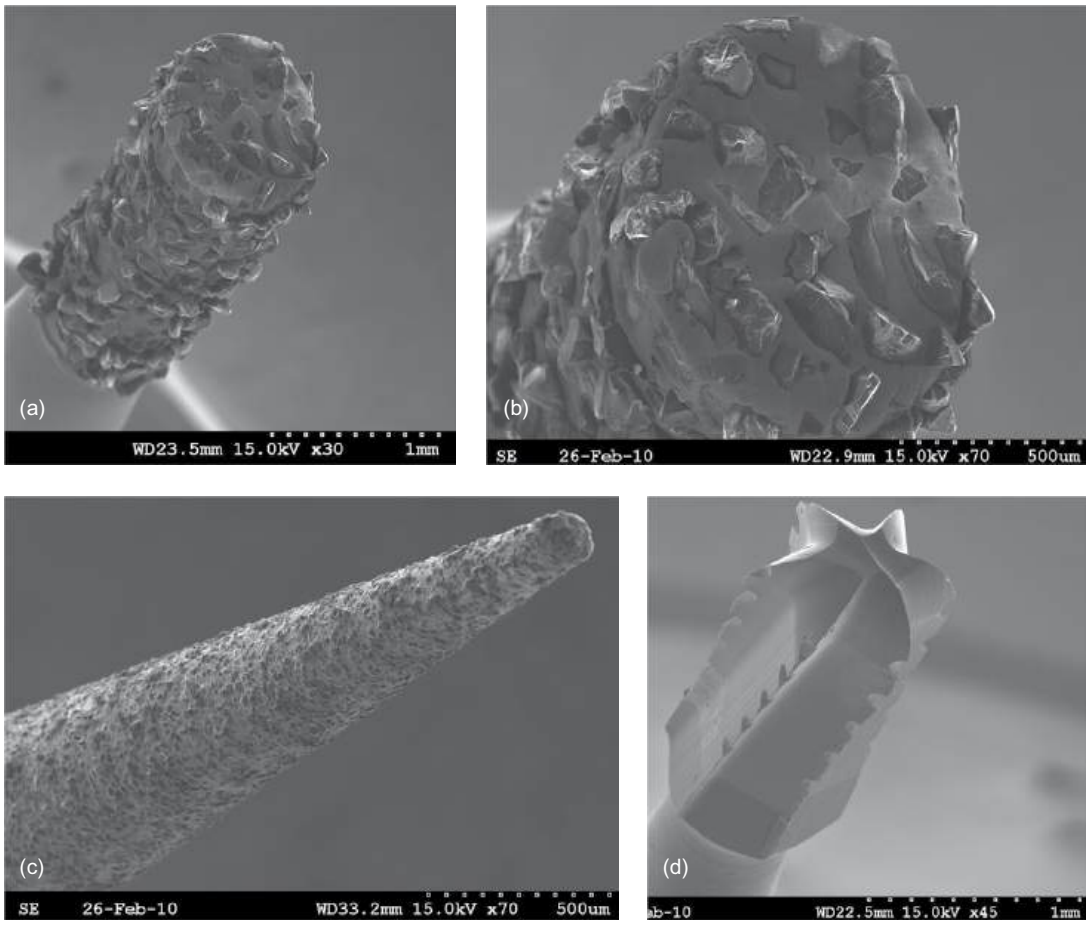


Figure 5.33 **(a)** Scanning electron micrograph (×30) of a coarse-grit diamond bur. **(b)** The same bur as in **(a)** at 70× magnification showing the random, irregular shapes of the diamond grit. **(c)** 70× magnification of the yellow stripe diamond composite finishing bur with a much smaller grit particle size than the bur in **(a)** or **(b)**. **(d)** 45× magnification of a tungsten carbide Beaver bur used for cutting amalgam and gold. Note the sharp notches cut into each flute.

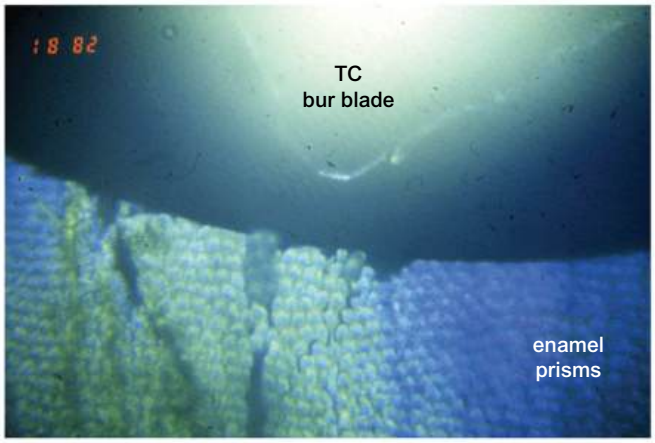


Figure 5.34 Cracks/clefts /cleaving of prisms (horseshoe-shaped structures) after enamel has been cut using a tungsten carbide (TC) bur in an air-turbine handpiece. This inherent damage is unavoidable clinically, but can be minimized by using only gentle pressure when operating, copious water spray, and new, sharp burs in a handpiece with sound bearings, allowing vibration-free rotation. The surface and subsurface damage will have implications for adhesive bonding, as the shrinkage stresses of adhesive materials exerted on these weakened prisms may open up the cracks, further leading to cohesive failure (field width 200 µm).

rule, hard materials (sound enamel, cast metals, and restorations) are cut more efficiently at high speeds and less pressure, whereas softer material (caries) is removed more efficiently at lower speeds and higher torque.

- Water spray ejected from tiny nozzles built into the handpieces should be directed to the tip of the rotating bur head (see Figure 5.27). This cools the tooth surface as it is cut, so preventing excessive frictional heat generation from having a deleterious effect on the pulp. It also helps to flush away debris from the cut tooth surface and the bur head itself. The dental nurse should endeavour to keep the mouth mirror free from spray during rotary instrument use, using the 3-1 air/water syringe and aspiration.
- Some handpieces, when coupled to the main dental unit enabled with fibre optics, have the ability to transmit the fibre-optic light from the head of the handpiece directly on to the bur head (see Figure 5.28). This can greatly improve direct vision.
- Air-turbine, rotary burs produce less vibration and pressure. Latch-grip burs have the potential to vibrate.
- The cutting action of rotary (and hand) instrumentation causes a *smear layer* to be formed on cut hard tissue surfaces (usually < 100 µm thick). This is a tenacious layer of organic, inorganic, and bacterial debris which is either removed or incorporated in a modified form into the bond when placing an adhesive restoration (see Chapter 7, Section 7.2.4).

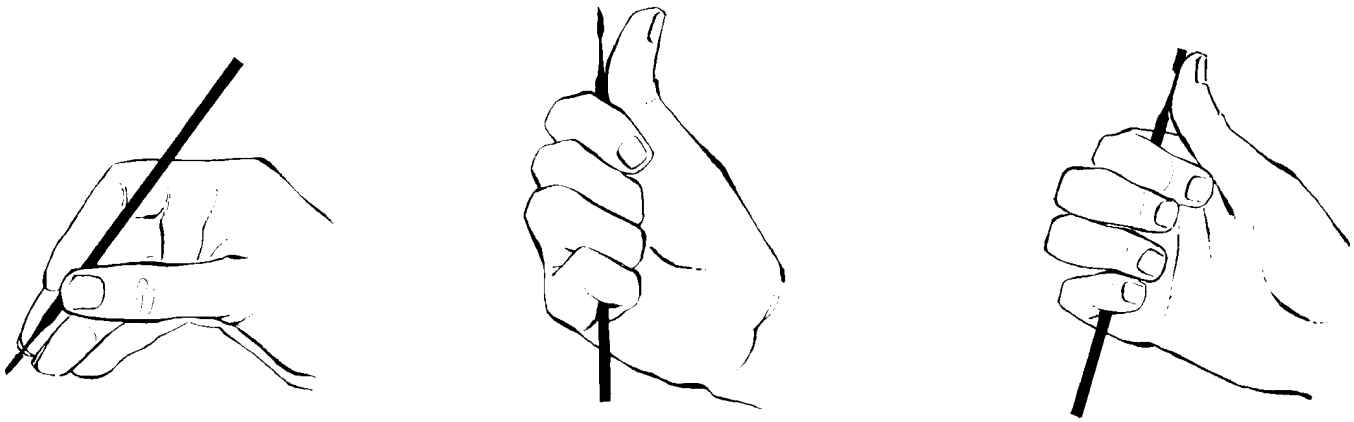


Figure 5.35 From left to right: pen (most common, most control for fine movement), palm, and finger grip of hand instruments. The latter two have a more limited use for more forceful movements in the maxillary arch. Instrument is held between the thumb (for support) and forefinger with the handle across the palm, clasped by the remaining fingers.



Figure 5.36 Examples of finger rests used by a right-handed dentist while operating a dental handpiece. On the left is a finger rest on the mandibular incisors when operating in the mandibular left quadrant. The middle picture depicts the finger rest on the mandibular right lateral incisor/canine when operating on the mandibular right quadrant. The right image shows a finger rest on the maxillary right canine/premolar region when accessing the maxillary anterior teeth.

- Finger rests: support required for both hand and rotary instrumentation can be achieved by resting the free fingers of the operating hand on a firm structure in the patient's mouth (i.e. adjacent/contralateral/opposing teeth). This provides fine control and reduces the risk of iatrogenic damage, especially if the patient moves suddenly and unexpectedly (see Figures 5.35 and 5.36).
- The free hand is used to hold the mouth mirror to provide indirect vision of the maxillary arch, reflect light intra-orally, and retract soft tissues.
- For the inexperienced operator using rotary instruments (and even some experienced ones!), there is a risk of damaging adjacent sound teeth/restorations (operator-induced, *iatrogenic* collateral damage). As a novice, there may be a case for protecting the adjacent tooth surface with a matrix band when operating proximally, but this then has the disadvantage of interfering with vision.
- With respect to caries removal, burs are not self-limiting, and rely on the operator to distinguish between tissue that requires excavation and that which does not. There has been research into the development of a plastic, single-use bur (see Figure 5.29) which only removes tooth structure that is softer than itself (anything harder and it will blunt during use) and so is more self-limiting. Hand excavators rely on the tactile feedback offered to the operator to distinguish the tissue requiring excavation and permit finer discrimination between the carious dentine zones (infected vs. affected vs. sound; see Chapter 1) than rotary instrumentation.

5.8.4 Dental air abrasion

- Developed in 1945 by R. B. Black, air abrasion is a pseudo-mechanical method for cutting hard dental tissue where the tooth surface is bombarded with high-velocity dry abrasive particles in air, transferring kinetic energy to the tooth surface, which is micro-chipped away.
- Adjustable parameters include the air pressure, powder flow rate, operating distance, type/size/morphology of the abrasive particle, and diameter of the nozzle tip, which all affect the efficacy and rate of cutting.
- Particles approved by the US Food and Drug Administration (FDA) or with CE accreditation for clinical use in the mouth include > 27 µm alumina and bioactive glass powders.
- Useful for minimally invasive operative/reparative dentistry with no heat, vibration, pressure, pain, or noise generated—rapid operator-dependent tissue removal with alumina particles. The extent of surface/subsurface hard tissue damage (cracks/cleft formation) is far less than when using rotary instruments, so making the air-abraded hard surface more suitable for adhesive bonding (see Figure 5.37).



Figure 5.37 An air-abrasion handpiece being used to prepare the labial enamel surfaces of the upper anterior teeth for direct resin composite veneers. Powder used: dry, 27 µm alumina. Note the blue aspirator tip held close to the operating field to vacuum up the spent powder.

Q5.6: What clinical situations may benefit from using air-abrasion tooth preparation?

- Bioactive glass particles can be used for extrinsic stain removal, desensitization of exposed cervical dentine, composite removal, and selective demineralized enamel removal (e.g. preventive resin restorations in high caries risk patients, cleaning residual resin cements after orthodontic bracket removal). Research also indicates a potential use of bioactive glass powders to aid remineralization of white spot lesions.
- New powders are being investigated for self-limiting carious dentine removal and their remineralization potential.
- Multi-chambered air-abrasion units allow a metered flow of different particles (alumina, bioactive glass, sodium bicarbonate) in a shroud of water, reducing the dust generated by the process and permitting instantaneous initiation and termination of the abrasive stream (see Figure 5.38); rubber dam is advisable for patient comfort.



Figure 5.38 An example of a two-chambered dry/wet cutting air-abrasion unit.

5.8.5 Chemo-mechanical methods of caries removal: Carisolv™ gel

- This is a gel-based dentine caries removal system that reacts with carious dentine that has undergone proteolytic breakdown of collagen, causing further collapse of the collagen network for easy final removal with curetting hand instruments or ceramic/plastic burs in a speed-reducing handpiece. This chemistry confers an element of self-limiting caries removal ability.
- Consists of a 0.1% hypochlorite-based alkaline gel (pH 11) with amino acids preventing breakdown of sound collagen fibrils.
- Stainless steel abrasive, non-cutting hand instruments permit carious dentine removal from cavities (see Figure 5.39). The system requires cavitated lesions for direct access to the carious dentine. Ceramic or plastic burs can also be used to aid excavation.
- As excavation is concentrated within caries-infected dentine, patients do not always require local anaesthesia. Gentle hand instrumentation pressure (the same force as normal tooth brushing) contributes to improved patient comfort (see Figure 5.40).



Figure 5.39 A selection of non-cutting, abrasive hand instruments for use with Carisolv™ gel.

5.8.6 Other instrumentation technologies

As can be seen from Table 5.5, there are several clinical methods for cutting teeth and removing caries. Ultrasonic and sonic instrumentation use the principle of probe tip oscillation and micro-cavitation to chip away hard dental tissues. Lasers transfer high energy into the tooth through water, causing photo-ablation of hard tissues. Great control is required by the operator in order to harness this energy effectively, and the effects on the remaining enamel, dentine, and pulp are under investigation in terms of residual strength and adhesive bonding capabilities. Enzymatic (including pepsin-based and papain-based) solutions are being investigated to help further breakdown of collagen in already softened carious dentine in the hope of developing a more self-limiting technique for removing caries-infected dentine alone. Other chemical methods include photo-activated disinfection (PAD)—introducing tolonium chloride into the cavity, which is taken up by the remaining bacteria in the cavity walls and then activated using light of a specific wavelength, so causing cell lysis and death—and ozone (gaseous ozone infused into early lesions, causing bacterial death). These technologies currently suffer from a lack of clinical research to validate them for routine evidence-based clinical use.

It is clear then that minimally invasive operative cutting technologies must aim to preserve as much naturally repairable tooth structure as possible, and at present no perfect instrument exists. Therefore the operator's knowledge of the histology of the dental substrate must be combined with their understanding of adhesion chemistry and the clinical handling of the restorative materials and instruments, if they are to operatively manage caries successfully.



Figure 5.40 Buccal cavitated carious lesion (mICDAS 4) on the LR3. **(a)** Clear Carisolv™ gel sitting in the cavity for 30 s prior to mechanical agitation. **(b)** Gel and carious dentine agitated with a mace tip hand instrument using tooth-brushing force. Note how the caries-infected dentine 'emulsifies' into the gel, making it cloudy. A repeat application of gel was required in this case to obtain suitable carious dentine removal. No local anaesthetic or rubber dam was required. **(c)** Final cavities before restoration. There has been no direct exposure of the pulp, even though caries excavation is deep. Caries-affected dentine (scratchy but sticky/flaky) has been retained around the gingival margins in this case, along with an intact periphery of enamel (prepared using chisels/gingival margin trimmers; see Figure 5.25). (See also Section 5.9.) **(d)** Cavity has been restored provisionally with a glass ionomer cement (GIC), during the stabilization phase of the care plan (see Chapter 3).

Q5.7: Look at Figure 5.40. Why was the decision taken to retain such relatively large quantities of caries-affected dentine, especially at the gingival periphery of this particular lesion?

5.9 Minimally invasive operative management of the carious lesion

5.9.1 Rationale

The vital aspects of prevention/control of disease and patient management have been discussed in Chapters 3 and 4. There are important factors that will influence the decision to intervene operatively to treat a carious lesion:

- *Alleviate pain/'protect' the pulp*: caries removal and a sedative dressing/definitive sealed restoration can remove the symptoms of an acute, reversible pulpitis and allow the dentine–pulp complex to react and heal.
- *Restore form, function, and appearance*: a large cavitated lesion will weaken the tooth, impeding normal mastication. Smaller unsightly lesions in the anterior aesthetic zone may require operative intervention, rather than preventive measures alone, for reasons of appearance affecting the patient's inclination to smile, and ultimately their self-confidence.
- *Aid plaque control*: rough/cavitated tooth surfaces may be difficult for the high-risk patient to keep plaque-free (especially in deep occlusal pits/fissures or proximally).
- *The patient's caries risk assessment/prediction*: relatively small, early lesions in a high-risk patient with poor plaque control where repeated preventive management has been unsuccessful may require operative intervention, whereas in a lower-risk patient these same lesions may benefit from non-operative control measures (see Chapter 4). This is particularly relevant if previous preventive regimens have failed and lesions are progressing actively.

5.9.2 Minimally invasive dentistry

This is an essential part of the overall minimum intervention caries management philosophy followed in this book and dental practice—an approach where the oral healthcare team bases its individualized patient care on early detection of disease, risk assessment, diagnosis, and prevention/control of further disease with tailored recall appointment frequencies (see Chapters 3 and 9). When operative intervention is required for the reasons mentioned previously, the approach should be *minimally invasive*, that is:

1. excavation of the unrepairable, diseased enamel and dentine only, keeping cavities as small as possible and preserving sound and repairable tissue where possible
2. physically and chemically modifying/optimizing the remaining cavity walls in order to
3. restore cavities with suitable adhesive materials which will:
   - support and strengthen the remaining tooth structure
   - promote remineralization and potentially have antibacterial activity
   - seal off any remaining bacteria from their nutrient supply, thereby arresting the caries process in the residual caries retained within the cavity depths
   - restore appearance, form, and function with appropriate long-term success.

In order for this minimally invasive approach to be successful, an integration of your knowledge of histology with the chemistry and handling of dental materials is essential. In addition, it is vital that the patient understands their responsibility for controlling and preventing further disease progression, as discussed in Chapter 4. Remember that 'drilling and filling' teeth does not cure caries! Any restoration placed is only ever as good as the operator placing it and the patient looking after it. In modern dental practice, it is imperative that detailed written records are kept highlighting all of this.

The decision-making process and execution of minimally invasive caries excavation/cavity preparation can be divided into the stages described in the flowchart in Figure 5.41, and will be discussed in Section 5.9.3.

5.9.3 Enamel preparation

The aims of cutting through enamel when minimally invasively operatively managing a carious lesion are to:

1. gain visual/instrumentation access to the full extent of the deeper carious dentine requiring removal
2. remove demineralized, weakened (and often unsightly frosty white) carious enamel
3. create a sound peripheral enamel margin to which an adhesive restorative material can bond and form a seal.

Figure 5.42 shows a histological cross-section through a cavitated, mICDAS 4 lesion with unsupported and demineralized enamel, and the extent of enamel preparation required. These margins may be bevelled lightly to increase the surface area for adhesion. Figure 5.43 shows the clinical occlusal appearance of such enamel removal using a diamond or TC bur in an air-turbine handpiece. This cutting process must be undertaken with only gentle pressure, avoiding the use of coarse grit diamond burs, so as not to cause excessive chipping or cracking of the remaining structure (see Figure 5.34).

Direct visual/instrumentation access to occlusal, buccal, and lingual/palatal carious surfaces is relatively straightforward with good soft tissue retraction and intra-oral lighting. Posterior proximal surface lesions (mesial and distal) can be more awkward, and approaches here would include the following:

- Occlusally, accessing initially just medial to the affected marginal ridge and eventually sacrificing it, thereby creating a proximal box (see Chapter 8).
- Occlusally, accessing medial to the marginal ridge, directing the bur towards the proximal lesion, *tunnelling* beneath the marginal ridge, and conserving it. This is only successful clinically if the lesion is sufficiently small, and it might then be debated whether operative intervention is appropriate in the first place! There is also a high risk of fracture of the weakened marginal ridge enamel in service, following restoration with a suitably flowable adhesive restorative material.

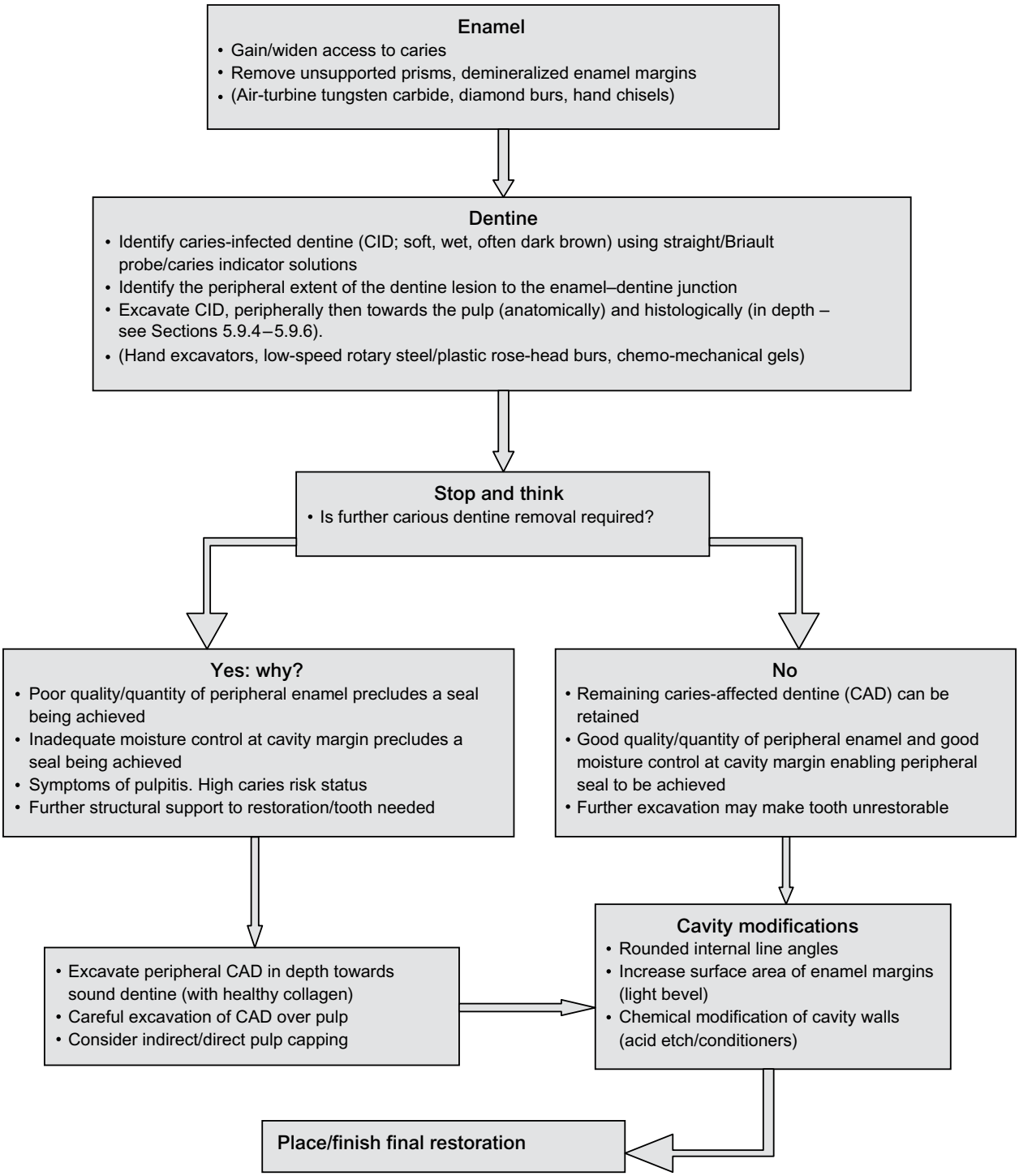


Figure 5.41 Flowchart showing the decision-making process and stages of execution of caries removal following the minimally invasive approach. These stages will be discussed in detail in Section 5.9.3.

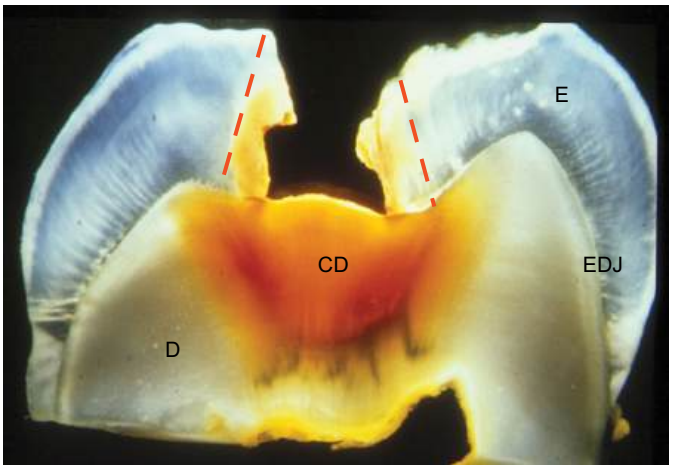


Figure 5.42 A thin mesio-distal longitudinal section through a cavitated lesion (mICDAS 4) viewed in reflected light. Note the demineralized, overhanging enamel bordering the cavity, with the undermining spread of the dentine lesion at the EDJ (see Chapter 1). The red broken lines indicate the extent of enamel removal required in order to achieve a sound enamel margin with supported prism structure, suitable for affording a seal with an adhesive restorative material while still permitting access to the underlying carious dentine. E, enamel; D, dentine; CD, carious dentine; EDJ, enamel–dentine junction.

Q5.8: Look at Figure 5.42. What is the name given to the appearance of the alternate light and dark striping evident in the inner third of enamel, arising from the EDJ and radiating towards the tooth surface?

- Buccally/lingually/palatally, especially if the tooth is rotated and has undergone some gingival recession. Tunnelling from these aspects can result in successful management as long as the lesion is not too extensive, and effective moisture and cavity margin control can be achieved.
- Directly mesial or distal, if the adjacent tooth is missing and space is present to permit direct visual/instrumentation access.

Anterior proximal lesions are approached conventionally from the lingual/palatal aspect to maintain the aesthetics of the natural labial enamel. However, minimally invasive tooth preservation principles would argue that a labial approach might be appropriate if this led to the least amount of healthy tooth tissue being destroyed. In the modern era of high-quality aesthetic direct adhesive restorative materials, this is an approach that can be encouraged, as good operator clinical skills can achieve a high-quality aesthetic result.

5.9.4 Carious dentine removal

When excavating dentine caries, careful consideration must be given to both of the following:

- 'anatomical extent' of the lesion (i.e. the lateral extent from the EDJ lesion periphery across to the caries overlying the pulp)

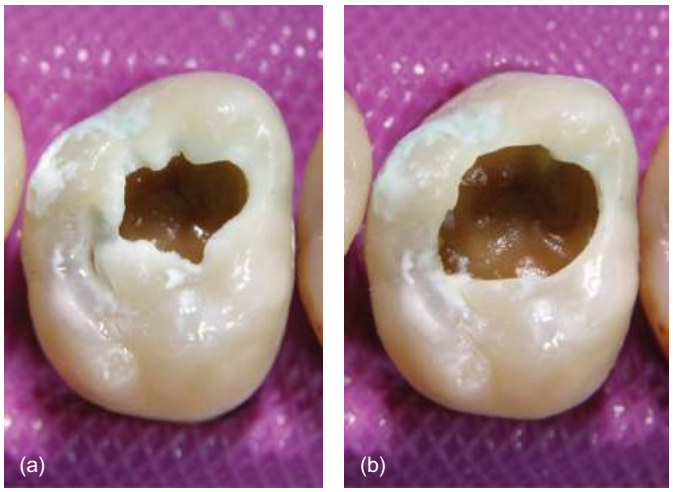


Figure 5.43 An occlusal cavitated carious lesion UR7 (mICDAS 4). **(a)** The lesion at presentation with frosty white demineralized, friable, and unsupported enamel at the cavity margin with carious dentine visible. **(b)** Enamel access has been widened by removal of the weakened peripheral enamel using a diamond bur in an air-turbine handpiece for < 5 s. The margin has been bevelled lightly.

- 'histological depth' of the lesion (i.e. the collagen and mineral content of caries-infected dentine vs. caries-affected dentine vs. sound dentine) (see Chapter 1).

Dentine caries removal must be tailored to the individual lesion, tooth, and patient. Following the minimally invasive approach, smaller cavities and restorations are created with the following benefits, ultimately leading to restorations with increased longevity and, more importantly, reduced tissue destruction and increased tooth longevity:

- Adhesive restorative materials are often easier to handle and place without voids in smaller quantities.
- Moisture control, cavity margin seal, and finish can be better regulated.
- The restored crown is often strengthened due to greater retention of natural, repairable tooth structure.
- It is simpler, and therefore results in improved patient care and maintenance.

5.9.5 Peripheral caries (EDJ)

Prevention of microleakage and subsequent recurrent caries at the cavity margin or caries progression beneath a restoration depends on the seal formed between the restoration and tooth structure at the periphery of the cavity.

- Histologically, the optimal peripheral seal achieved will be at the interface between the adhesive restorative material and sound enamel > sound dentine > caries-affected dentine > caries-infected dentine/carious enamel (the latter two being the least satisfactory and not recommended).
- The adhesive seal is created optimally between an adhesive restoration and histologically sound enamel. Therefore if sound enamel lines the entire periphery of the cavity, and moisture control is optimal (e.g.

an occlusal cavity), limited quantities of brown-discoloured, repairable, caries-affected dentine may be retained at the EDJ. However, this should be removed selectively, down to deeper sound dentine, if there is a risk that the brown discoloration may shine through the restoration, thus jeopardizing the aesthetic outcome (especially in the anterior aesthetic zone, canine–canine).

- If minimal/no enamel is present at the EDJ after peripheral caries removal, and adequate moisture control cannot be guaranteed (e.g. cervical or proximal lesion, base of a proximal box preparation close to the gingival margin), sound dentine is a prerequisite at the EDJ in an attempt to maximize the bond and adhesive seal between the peripheral dentine and final restoration. However, if this means extending the cavity a long way subgingivally, consideration must be given to the long-term success rate of the final restoration and ultimately the final restorability of the tooth in question (see Figure 5.40).
- Distinguishing between caries-infected, caries-affected, and sound dentine is a rather subjective skill, and the operator usually uses differential tactile judgement (see Table 5.7). Brown discoloration of the dentine at the EDJ is an indicator neither of its infectivity nor of the need for its removal (unless in the anterior aesthetic zone).
- In no clinical situation should caries-infected dentine be retained at the EDJ, as this dentine is essentially necrotic and cannot be adhered to or a seal achieved. It is also too soft to physically support a rigid restoration under occlusal load.
- Instruments commonly used include hand excavators or steel rosehead burs in a low-speed handpiece. Other technologies might include chemo-mechanical agents (see Table 5.5).

5.9.6 Caries overlying the pulp

When excavating dentine caries overlying the pulp, consideration must be given to its *proximity* to the pulp (see Section 5.11) and any *symptoms*.

- Where a peripheral seal can be achieved (enamel margins intact) and pulp symptoms are not present, quantities of caries-affected dentine may be retained overlying the pulp, thereby reducing the risk of pulp exposure. Clinical evidence now shows that affected dentine provides better indirect pulp protection than any artificial restorative material in this situation.
- If the dentine lesion has only penetrated into the middle third of dentine radiographically, thanks to tertiary dentine being laid down at the dentine–pulp border, excavation to sound dentine may be achievable to improve the adhesive bond and seal (especially if the enamel margins are not intact).
- Where chronic pulp symptoms persist along with radiographic changes at the apex, caries excavation needs to be complete, exposing the pulp chamber and its necrotic contents if necessary with a view to the continuation of endodontic treatment.

Table 5.6 summarizes the interlinked factors that affect the decision to remove/retain carious dentine. It must be understood that there is no absolute correct or incorrect amount of caries that should be excavated, but whatever decision is made by the operator, it should be made for *pragmatic*, *justifiable*, and *documented* reasons. Different quantities of caries can be removed or retained in different parts of the same

Table 5.6 A summary of the interrelating factors that affect the decision about how much dentine caries to excavate (infected vs. affected vs. sound)

Factors affecting amount of carious dentine removed	Comments 'GIC, glass ionomer cement'
Patient's caries risk	High-risk, uncontrolled caries progression, unsuccessful prevention regimes: lesions may be treated with excavation to sound dentine where possible. Important for operator and patient to appreciate the reduced longevity of the tooth–restoration complex due to the lack of patient adherence to preventive strategies (see Chapter 9)
Patient/oral factors	Limited oral opening, physical/mental disabilities can affect visual/instrument access directly/indirectly. Broken down/rotated/partially erupted teeth can hinder rubber dam placement. Sedation/general anaesthesia may be required
Pulp vitality (sensibility)	Chronic symptoms of an irreversible pulpitis will mean pulp extirpation after complete caries removal to sound dentine
Pulp proximity	If no/limited acute pulp symptoms, CAD retained over the pulpal aspect of cavity to allow remineralization from dentine–pulp complex—*indirect pulp capping/protection*
Remaining coronal tooth structure	Sufficient supragingival tooth structure must remain to support the restoration long term. Excavation to sound dentine should not compromise the physical/adhesive properties of the restorative material (e.g. deep peripheral excavation to sound dentine resulting in a cavity margin of > 2 mm subgingivally, compromising moisture control and marginal adaptation)
Material factors	Retention mechanisms—mechanical, micro-mechanical, chemical adhesion. An adverse oral environment can affect the set and physical properties of some materials (e.g. GICs in xerostomia)

CAD, caries-affected dentine.

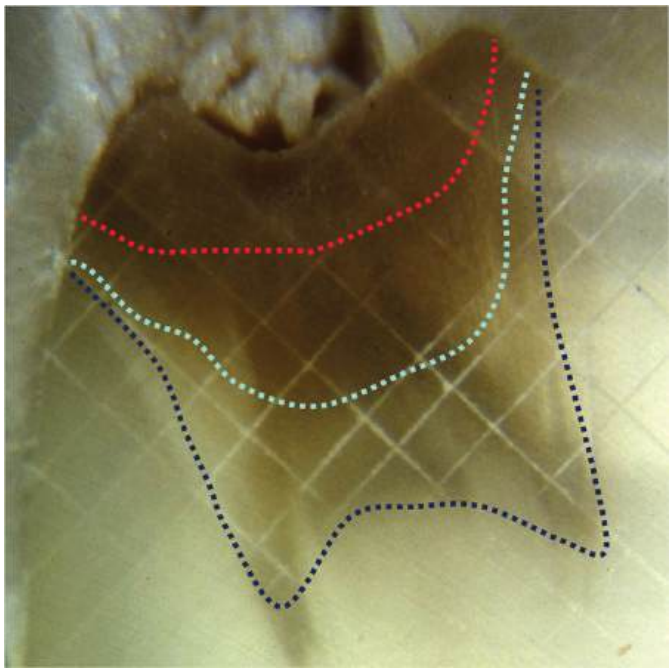


Figure 5.44 A cavitated dentine lesion (mICDAS 3/4) showing the colour gradation from the superficial caries-infected dentine subjacent to the EDJ through to the deeper levels of caries-affected dentine, including the translucent zone and sound dentine. Three positions have been indicated for potential caries excavation endpoints: white line—excavation to sound dentine at EDJ and pulpal aspect; green line—less invasive excavation retaining stained dentine at the EDJ and affected dentine over the pulp; blue line—minimally invasive excavation finishing on sound enamel at the EDJ and infected/affected dentine over the pulp. Note how all of the potential endpoints finish on sound enamel, so providing a sealed margin. None of these endpoints are right or wrong, but it is essential that the histology at each level is appreciated in order to enable a successful adhesive restoration to be placed.

Q5.9: Which of the three levels do you think would be best to stop caries removal in the lesion shown in Figure 5.44?

cavity. Figure 5.44 highlights this variation in dentine caries removal, showing three levels of potential excavation endpoint, depending on the combined outcome of the factors discussed in Table 5.6.

5.9.7 Distinguishing the zones of carious dentine

The clinical discrimination between caries-infected, caries-affected, and sound dentine is at present a subjective skill gained by using a combination of an understanding of caries histology and clinical experience. These boundaries are not clearly defined; the histological and bacterial changes occur throughout the whole lesion as a continuum from superficial to deeper layers, with different rates of progression within the individual lesion itself. Table 5.7 summarizes the techniques available—both clinical and through research development—to help to enable the operator to distinguish these important 'histological' layers. From the table it can be seen that none are truly objective, with interpretation required to decide the excavation endpoint and linked to the factors discussed in Table 5.6.

5.9.8 'Stepwise excavation' and the atraumatic restorative technique (ART)

These two operative techniques are the original and more modern application, respectively, of the minimally invasive biological approach to managing larger cavitated carious lesions. Both can use simple hand instrumentation (spoon-shaped excavators) to remove the necrotic, superficial layer of caries-infected dentine and some caries-affected dentine also if required (see Figure 5.45). The two stages of the original stepwise excavation procedure included the placement of a calcium hydroxide lining and a temporary zinc polycarboxylate cement restoration. Between 6 and 9 months later, this restoration was removed completely, exposing the remaining, now arrested caries-affected dentine (darker, harder, and dryer) and newly deposited tertiary dentine at the pulp–dentine border. The residual stained dentine was now removed



Figure 5.45 Hand excavator scooping through necrotic, soft caries-infected dentine once the enamel access has been widened (reference grid lines on the dentine face were placed for experimental purposes).

Table 5.7 Clinical and research methods used to help to discriminate clinically between the three histological zones of carious dentine (infected, affected, and sound). Note that these zones do not have distinct boundaries, but have a gradient of histological/bacterial change through them from the enamel–dentine junction (EDJ) to the pulp (see Chapter 1)

Discriminating methods		Caries-infected dentine	Caries-affected dentine	Sound dentine
Clinical	Visual	Often a colour gradient from (see Figure 5.44): Dark brown → paler brown/translucent → yellow/white Difficult to judge clinically/OK when examining a sectioned lesion (research)		
	Tactile	Soft/sticky feel with a sharp dental probe (straight or Briault)	Both sticky/flaky and scratchy feel	Scratchy feel to a sharp dental probe
	Caries detector dyes	'Fusayama' dyes—based on propylene-glycol, collagen-based stain. Attempts to discriminate infected vs. affected, but research has shown that these dyes stain deeper collagen and permeate into affected/sound dentine zones, often leading to cavity over-preparation		
Research	Bacterial dyes	Dyes under research development reacting to bacterial redox by-products, using their concentration gradient drop from infected through affected to sound dentine. Have potential to be zone-selective		
	Fluorescence	Research into the natural fluorescence of dentine caries using confocal fibre-optic microendoscopy/fluorescence lifetime imaging (FLIM) indicates a discriminating level of fluorescence of infected dentine. A technique that might be developed for *in-vivo* use in the future		

using a steel rose-head bur in a low-speed handpiece, and a final amalgam restoration was placed.

The more recently described single-step ART proposes restoring the cavity with chemically adhesive, high-viscosity glass ionomer cement (GIC), forming a better adaptive seal to the remaining tooth structure (see Chapter 7) after the use of simple hand excavators to gain access to and remove the caries-infected dentine. Subsequently, this provisional restoration does not require complete replacement as in the original stepwise excavation procedure, as the arrested caries-affected dentine is sealed off from its nutrient supply. However, occlusal resurfacing of the GIC may be necessary after 2 to 3 years due to its wear and partial dissolution. This can be accomplished by cutting back the exposed worn GIC surface by up to 2 mm in depth, and bonding a layer of resin composite on to the freshly exposed, mature GIC—an *adhesive layered*, *laminate* or '*sandwich*' *restoration* (see Chapters 7 and 8). Indeed, systematic reviews of partial/incomplete minimally invasive caries removal have highlighted how the one-stage procedure with no re-entry stage, that seals in residual affected dentine, results in significantly less pulp exposures than the original two-step stepwise excavation procedure. This finding was the result of an increase in pulp exposures in the second re-entry procedure where deeper dentine was removed (forcibly and unnecessarily) too close to the pulp.

Both stepwise excavation and ART follow the underpinning principles of minimally invasive dentistry. As this is a more accepted contemporary rationale for operative caries management, these terminologies are now becoming redundant as current adhesive dental materials are more capable of sealing in and potentially rehabilitating diseased tissue.

5.10 Cavity modification

Once caries removal has been achieved, the dentist needs to stop, think, and decide which restorative material to use, and then modify the remaining cavity accordingly, if required (see Figure 5.46). These changes may be necessary to improve the retentive properties or to maximize the material properties (e.g. bulk/margin strength or fracture resistance).

- *Retention*: the property of a cavity that resists displacement of the restoration in the direction of its insertion.
- *Cavity support*: the cavity property that prevents displacement (or fracture) of the restoration in any other direction, including internal dislodgement within the cavity itself. This feature relates to the morphology of cavity walls/floors and rounded internal line angles.

Table 5.8 outlines the different macro- (bur-induced) or micro-modifications (chemically induced) dependent on the restorative material to be used.

Table 5.8 Potential cavity modifications required after caries removal, depending on the material chosen to restore the cavity*

Restorative material	Cavity modification		Comments
Resin composite/GIC	macro	Enamel margin bevel (short/long)	Removes grossly unsupported prisms Increases surface area for bond/seal Provides a sound enamel surface to facilitate optimal adhesive bonding (see Figures 5.42, 5.43, and 5.46b)
All	macro	Rounded internal line angles	Reduces both internal stresses and risk of crack propagation within the restoration (see Figures 5.46, 5.47, 5.48, 5.49, 5.50)
Resin composite	micro	Enamel acid etch	37% orthophosphoric acid removes smear layer, and selectively demineralizes prisms, creating micro-mechanical undercuts for resin retention
GIC	micro	Dentine conditioner	10% polyacrylic/citric acids modify or remove smear layer, preparing the surface for chemical adhesion (Ca^{2+})
Amalgam	macro	Cavity undercuts, grooves, slots, flat surfaces	Cavities with wider bases than orifices are required for retention of amalgam—undercuts. Slots/grooves help to prevent further displacement of the restoration (see Figures 5.48 and 5.49) Flat cavity surfaces with rounded internal line angles help to improve the internal cavity support (see Figure 5.50)
All	macro	Pulp chamber	In endodontically treated teeth, the pulp chamber is used to add bulk and retentive features/increased surface area for the coronal restoration
Amalgam	macro	Nayyar core	Packing the coronally flared 2–3 mm of endodontically treated root canals to improve core retention
Gold, porcelain, resin composite (indirect restorations)	macro	Long marginal taper	Divergent tapered margins (7–10°) used to improve retentive features of indirect restorations (see Chapter 6)

*The modifications have been classified as macro (those created using a bur) or micro (those created using chemicals). See Chapter 7 for a description of the material science and Chapter 8 for practical considerations.

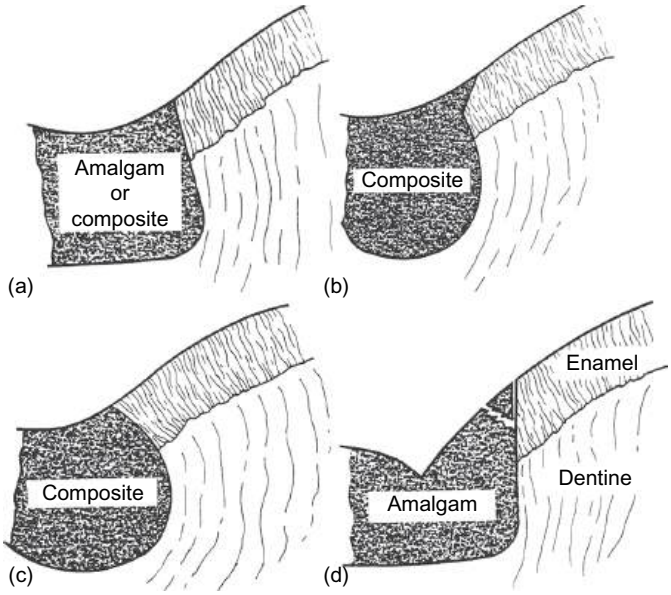


Figure 5.46 Diagrams outlining the different restoration margin angles and cavo-surface angles that are suited to different restorative materials. **(a)** Amalgam and resin composite margins require bulk for strength, so 90° restoration margin angles and cavo-surface angles will ensure sufficient intrinsic strength of both the restoration margin and cavity edge. **(b)** Resin composites adhere and support enamel to a degree, so light bevelling of the enamel surface can increase the surface area for adhesive bonding while also removing any potentially unsupported enamel prisms at the cavity edge **(c)**. **(d)** If the amalgam margin angle is too acute, the amalgam margin may be weakened sufficiently for marginal fracture to occur under occlusal loading. Note the internal rounded line angles in all of the cavities depicted.

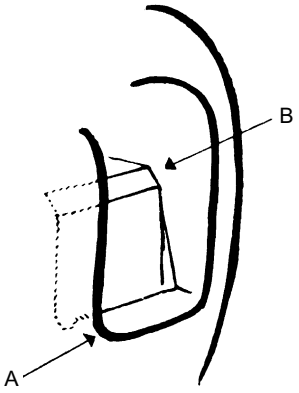


Figure 5.47 Diagram indicating the internal bevelling/rounding off of the internal line angles between adjacent internal cavity surfaces (A and B). This reduces the risk of crack propagation within the restorative material.

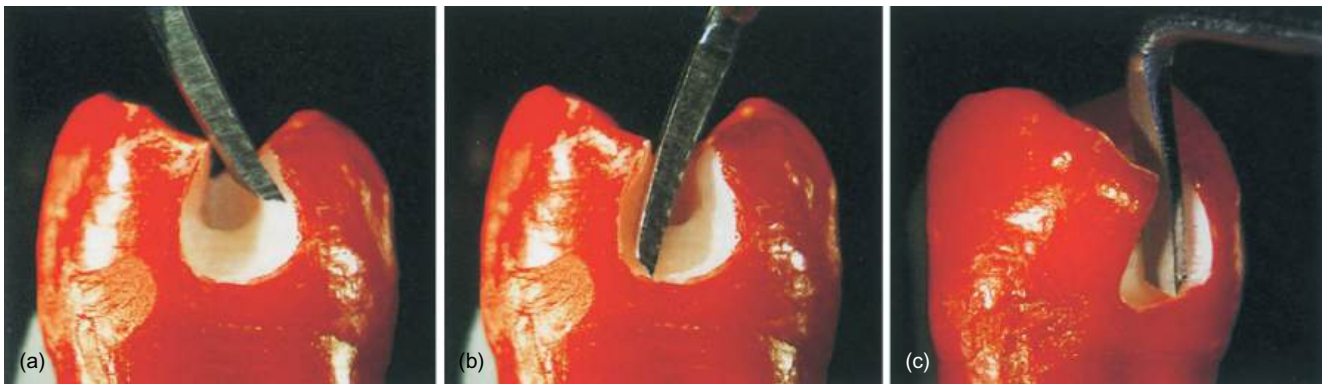


Figure 5.48 **(a–c)** Proximal box cavity prepared with undercuts (a wider base and a narrower occlusal opening) to prevent displacement of the restoration (amalgam) occlusally from its path of insertion.

Q5.10: Which hand instrument is shown in Figure 5.48?

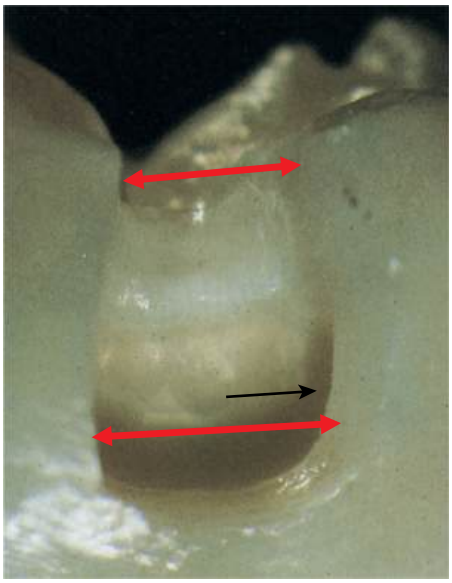


Figure 5.49 A proximal box cavity prepared with occlusal undercuts (a wider base and a narrow occlusal opening—thick red arrows) to prevent displacement of the restoration (amalgam) from its path of insertion occlusally. Note the rounded internal line angles. The thin black arrow shows the placement of a groove at the junction between the buccal wall of the box and the pulpal wall, so providing stability (resistance) to proximal displacement.

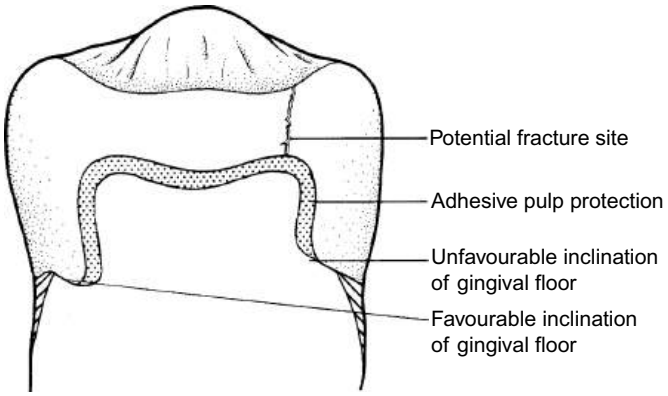


Figure 5.50 Potential problems of the morphology of cavity walls and floor, requiring modification. Rounded internal line angles reduce the risk of crack propagation and eventual restoration fracture of rigid materials (e.g. amalgam). Use of a rose-head bur to remove carious dentine tends to create the cavity floor morphology on the left, but care must be taken to remove any very prominent unsupported enamel lip at the periphery. The floor morphology on the right offers limited support to the final restoration and should be avoided.

5.11 Pulp protection

5.11.1 Rationale

The final and most important consideration along with any cavity modifications, prior to placing the restoration, is the long-term pulp status. This is linked to pulp *signs and symptoms* and *cavity proximity* to the pulp. Assuming that the symptoms and signs of pulp sensibility indicate a histologically viable pulp (see Chapter 3), the pulp tissues may require protection from:

- bacteria/toxins
- chemicals leaching from the restorative materials (e.g. unconverted resin monomers from resin composite restoratives, acid etchant)
- thermal and/or electrical stimulation via conductance through the overlying restoration.

If the floor of the cavity is close to the pulp but the pulp chamber has not been breached during minimally invasive caries removal, this protection can be afforded by a process termed *indirect pulp capping (IPC)*. If a small breach, or *pulp exposure*, has occurred (usually no wider than the tip of a Williams periodontal probe), created either iatrogenically or by the caries process, a *direct pulp cap (DPC)* may be placed in an attempt to maintain pulp viability, but the patient must be warned that this is not guaranteed, especially when this is a consequence of a deep carious lesion (see Figure 5.51).

5.11.2 Terminology

There are two older terms used in the dental literature to describe this form of material-based pulp protection where an exposure has not occurred—*cavity lining* and *structural base*. These terms were coined when amalgam was the only material of choice to restore cavities long term and it was thought that the pulp required an 'insulating' layer between it and the metal-based restoration. The excessively large, undercut cavities designed for amalgam could be reduced in size artificially by placing zinc oxide-based plastic materials as 'structural bases', on to which was packed a reduced volume of amalgam. In the contemporary era of minimally invasive operative dentistry, these terms should no longer be used, so as to avoid confusion.

5.11.3 Materials

Materials used for both IPC and DPC should ideally:

- be bactericidal (able to kill bacteria)/bacteriostatic (able to prevent them from multiplying)
- be mildly irritant to the pulp, to stimulate tertiary dentine bridge formation (usually via pH changes)
- be adhesive in order to effect a bond and seal
- not dissolve away over time
- be easily applied and strong in thin section

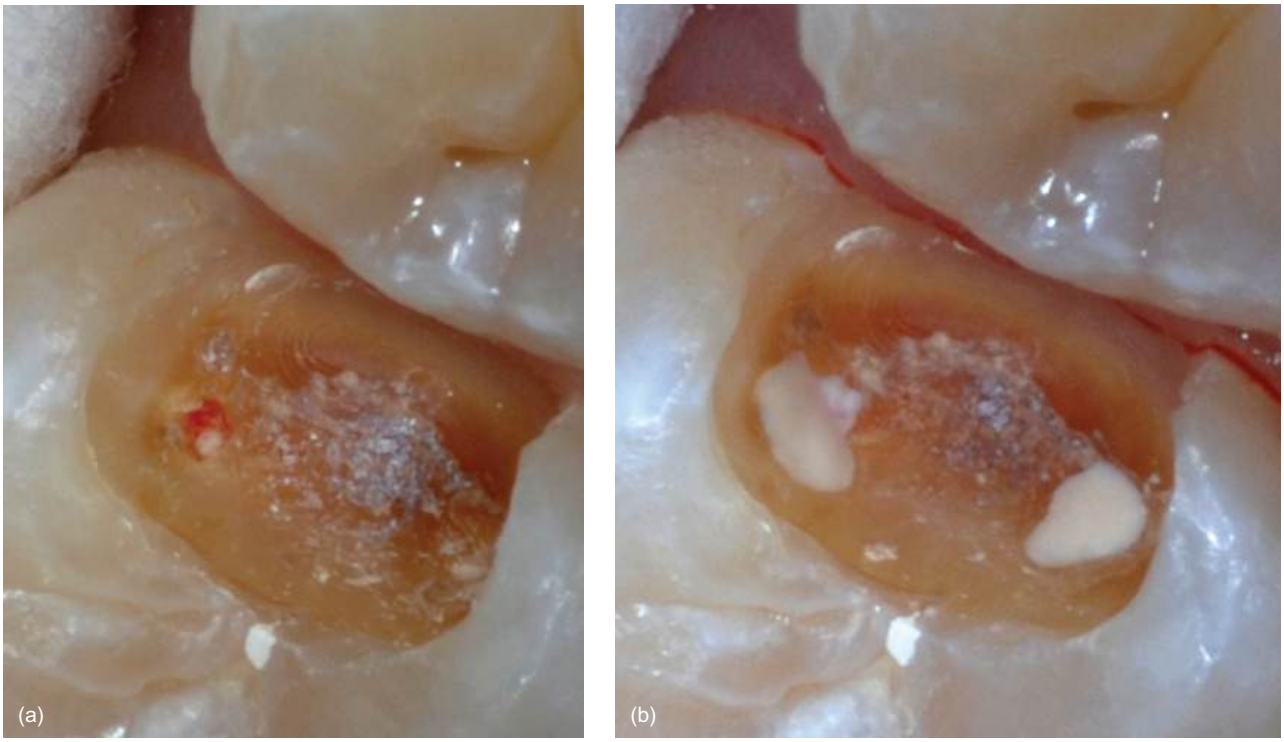


Figure 5.51 **(a)** Caries excavated from a proximal lesion in a vital tooth with residual caries-affected dentine evident on the pulpal aspect of the cavity. Note the spot of blood exuding from a small carious exposure through the pulp horn. **(b)** Setting calcium hydroxide has been placed over the small pulp exposures. (Courtesy of L. Mackenzie.)

Q5.11: After placing the direct pulp cap as shown in Figure 5.51, what are the next steps in the restoration of this cavity?

- be able to infiltrate ionically into the remaining dentine overlying the pulp, thereby strengthening/reinforcing it
- be biocompatible with the pulp and the overlying restorative material.

Clinical dental materials used for IPC and DPC (see Chapter 7) include:

- glass ionomer cements
- dentine bonding agents
- setting calcium hydroxide
- tricalcium silicate cements (mineralized trioxide aggregate (MTA), Biodentine™).

The first two materials in the previous list fulfil the majority of the criteria for an ideal IPC mentioned earlier. When using these adhesive materials as the final restorative solution, it can be argued that a separate thin layer of pulp protection is not necessary, as the properties are 'built in' to the bulk restorative material itself. Setting calcium hydroxide cements have been used for many years as pulp protection beneath amalgam restorations, but in recent times much more sparingly. The authors cannot suggest any positive indications for their use beneath adhesive restorations as an IPC. GICs and resin-based cements are being used increasingly in bonded amalgam restorations, so again the use of calcium hydroxide is ever more limited in this regard. As a DPC, the alkaline pH of calcium hydroxide still has a use as an inflammatory stimulant for pulp odontoblasts to produce tertiary dentine and so close the exposure. Unset MTA is primarily calcium oxide in the form of tricalcium silicate, dicalcium silicate, and tricalcium aluminate, with bismuth oxide added for radio-opacity. Several studies indicate its potential use as a DPC, as it increases the concentration of available calcium hydroxide, the primary reaction product between MTA and water, and provides a seal. However, its high solubility, long setting time (up to 3 hours), and expense are disadvantages. Both setting calcium hydroxide cements and MTA need a second protective covering prior to placing the final restoration. This is conventionally a GIC/resin-modified (RM)-GIC material (see Chapter 7). A faster-setting calcium tri-silicate restorative cement, Biodentine™, now exists, that incorporates the pulp protection properties within its bulk restoration (see Chapter 7).

Interestingly, clinical evidence now shows that retained caries-affected dentine overlying acutely inflamed pulps provides the optimum qualities of indirect pulp protection as long as the overlying adhesive restoration affords an adequate seal and the patient can maintain adequate oral hygiene and other preventive measures.

5.12 Dental matrices

(See also Chapter 8.) Cavities created with missing proximal walls present a technical problem when it comes to placing a direct plastic restorative material, as there is nothing to contain the restoration within the cavity or to prevent poor marginal adaptation (ledges, overhangs). *Circumferential* matrix bands, with retainers (e.g. reusable Siqveland and Tofflemire, disposable AutoMatrix and Omni-matrix; see Figure 5.52a) or without (e.g. copper rings, AutoMatrix or SuperMat matrices; see Figure 5.52b), or *sectional* matrix bands have been developed to rectify this problem (Palodent, Composi-Tight, V-Ring, QuickMat, among others). Traditional dental matrix systems consist of two components:

- *Single-use matrix band*: thin metal or clear plastic band of varying widths and gauges (30–50 µm) used to form the missing wall of the cavity. More rigid and wider bands used posteriorly (metal), and narrower, pliable, transparent plastic bands used anteriorly (for resin composites).
- *Retainer*: a device used to tighten and hold circumferential matrix bands in place proximally. These can be sterilized and used again. Sectional matrix systems have oval/round ring retainers with curved tines to adapt the matrix to the proximal tooth contour (see Figure 5.53).

5.12.1 Clinical tips

- Cervical margin adaptation of the band to the tooth is essential to prevent ledges or overhangs of excess restorative material, often requiring the use of wooden/plastic wedges in the interproximal space.
- Tight interdental contact points may need to be opened slightly by causing differential movement of the adjacent teeth within their periodontal ligament spaces. This is accomplished by pre-wedging using wooden wedges for a few minutes prior to placing the band.
- Sectional matrices may be of benefit in situations where teeth are rotated or tilted, thus making the placement of circumferential bands more difficult.

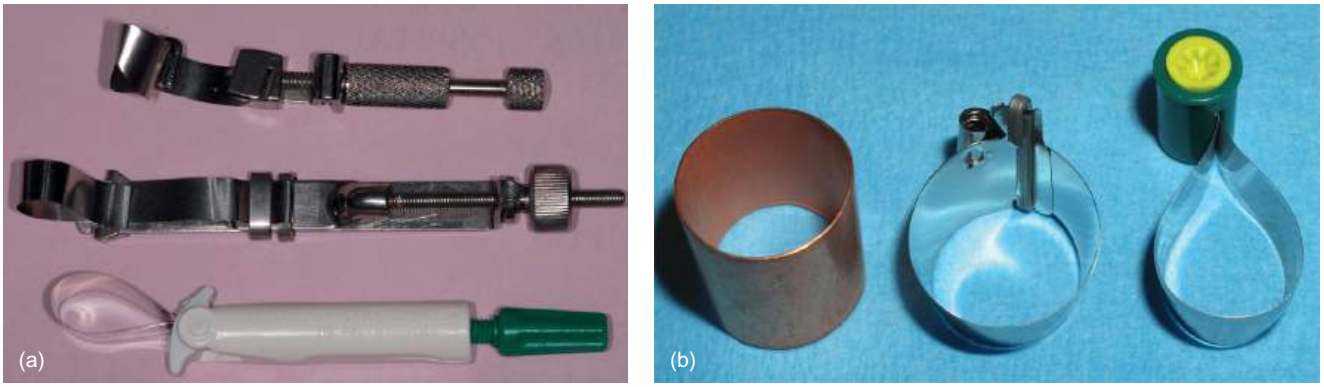


Figure 5.52 **(a)** Three examples of circumferential metal matrix bands with retainers. Top: Tofflemire retainer with precurved matrix band; middle: Siqveland retainer with matrix band; bottom: Omni-matrix disposable system. **(b)** Three examples of circumferential matrix bands without retainers: copper ring, AutoMatrix®, and SuperMat® systems.

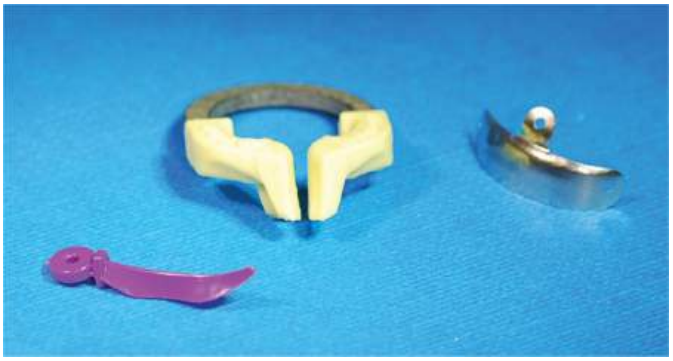


Figure 5.53 A precurved and contoured single-use sectional metal matrix band (right) with a plastic gingivally contoured wedge (left) and the oval ring retainer with curved tines (middle) to adapt the band to the approximal surface of the tooth. The retainer can be sterilized.

- Sectional bands are more adaptable to gain well-contoured proximal surfaces and contact areas.
- Circumferential bands can be more painful for the patient to endure than equivalent sectional bands.
- Clear plastic bands/wedges can be used when placing resin composite restorations to permit light penetration to the depths of the proximal cavity when curing the material.
- Rubber dam clamps may have to be removed prior to placing matrix bands.
- Anteriorly, if contact areas are tight and the flexible, clear plastic matrix band refuses to pass through incisally, the corner of the band can be cut to a point and passed cervically beneath the contact and then lifted incisally through it.

5.13 Temporary (intermediate) restorations

5.13.1 Definitions

A *temporary restoration* is one that is placed for short-term use (usually days or weeks), often between appointments in a course of ongoing definitive treatment (e.g. caries stabilization, multi-visit root canal treatment, simple fixed prosthodontics).

A *provisional restoration* is one that is placed for short/medium-term duration (weeks or months) where diagnostic value is gained from placing the restoration (e.g. complex fixed prosthodontic treatment when reorganizing a patient's occlusion or adjusting the vertical dimension of the patient's occlusion (see later); stabilizing caries to negate symptoms).

The dental material science of temporary restorations (GIC, zinc oxide eugenol, and zinc polycarboxylates) is covered in Chapter 7. Eugenol-containing temporary restorations should be avoided in cavities where the final restoration to be placed is a resin composite, as the eugenol adversely affects the resin's polymerization chemistry.

5.13.2 Clinical tips

- Care must be taken when placing temporary restorations to replicate the occlusion and marginal adaptation of the restoration.
- Zinc oxide-based materials have a relatively poor appearance, so in the anterior aesthetic zone the use of GIC-based materials might be considered.
- Patients must be made fully aware of the reason for and function of the temporary restoration.
- Adequate time must be allocated in the clinical appointment to place temporary restorations. Poor-quality marginal adaptation of temporaries can lead to further marginal staining, plaque accumulation, and gingival inflammation/bleeding which in turn often worsens the patient's oral hygiene procedures.
- Frustratingly, if temporary/provisional restorations have optimal aesthetics, the patient may not wish to have them replaced with their definitive counterparts!

5.14 Principles of dental occlusion

In order to successfully restore dental function in the long term, an understanding of human dental occlusion is of primary importance. This subject is complex, and many textbooks exist that discuss these complexities and their clinical relevance. The occlusal contacts between the maxillary and mandibular dentition are affected by the skeletal base relationship (condylar head vs. slope of the articular eminence, glenoid fossa), development of the maxilla and mandible, and direct factors influencing the development, position, and shape of the teeth themselves. Analysis of occlusion is critical in both the *conformative* and *reorganizational* approaches to managing dental care.

5.14.1 Definitions

Conformative approach

This approach to planning and placing restorations ensures that the pre-existing occlusal relationships are not altered in any of the three planes (antero-posterior, vertical, and horizontal). It is essential therefore to analyse aspects of the occlusion prior to commencing, so that the final restorations fit into the existing scheme. This approach most often applies when placing single or multiple plastic, direct restorations.

Reorganizational approach

This approach involves changes in occlusion, and is often undertaken when rehabilitating major occlusal discrepancies caused by tilting or over-eruption of teeth resulting in premature occlusal interference contacts and/or distortion of the occlusal planes causing reduced space into which to place definitive restorations. Here again considerable planning with the use of provisional restorations is required in order to calculate the tolerances of changing the occlusion. This approach is usually indicated in complex cases often involving the use of indirect restorations (crowns, bridges, dentures, or implants), and is beyond the scope of this particular textbook.

5.14.2 Terminology

The commonly used terminology and recorded positions of occlusal analysis are described in Table 5.9. QR code image 5.2 will take you to a glossary of prosthodontic terms.

QR code image 5.2. Scan this code with your mobile device to access a glossary of prosthodontic terms.

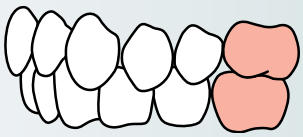
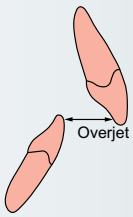
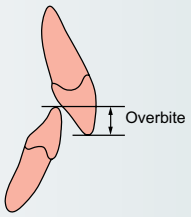
www.thejpd.org/article/S0022-39(13)0500175-7/fulltext

5.14.3 Occlusal registration techniques

The various static and dynamic relationships of the patient's occlusion should be analysed before starting any restorative procedure in the conformative approach to dental care provision, as well as checking again after the restoration(s) has been completed. There are many different clinical techniques available for checking direct occlusal contact relationships in the patient's mouth:

- *Articulating paper*: thickness in the range 40–200 µm; different colours that show up contact areas between cusp inclines and fossae on occlusion (see Figure 5.54).

Table 5.9 Various occlusal relationships, their definitions, and their relevance to restorative dentistry

Occlusal relationship	Definition	Diagram/comments
Intercuspal position (ICP) (static)	The occlusal relationship where there is maximum cusp interdigitation between the maxillary and mandibular teeth	Not always clinically reproducible, and can be affected by occlusal interferences 
Occlusal vertical dimension (OVD) (static)	The vertical relationship between the maxilla and mandible with the teeth in ICP	Can be reduced by tooth wear, missing teeth Can be compensated biologically by over-eruption of teeth/dento-alveolar compensation
Rest position (RP) (static)	The vertical relationship between the maxilla and mandible when resting comfortably in an upright position with relaxed facial musculature	Achieved after swallowing/yawning. Teeth usually slightly apart
Freeway space (FS) (static)	The vertical dimension difference between RP and OVD (ranges 2–4 mm)	Adaptive in the dentate patient Essential consideration for the restoration of the edentate individual with complete dentures
Overjet (OJ) (static)	The horizontal distance between the maxillary and mandibular incisal edges in ICP	
Overbite (OB) (static)	The vertical distance between the maxillary and mandibular incisal edges in ICP No vertical overlap can lead to an edge–edge incisal relationship or an anterior open bite (AOB)	
Retruded contact position (RCP) (dynamic)	The most retruded position of the mandible when there is initial cusp contact (described as the relationship where the condylar heads are rearmost within their glenoid fossae and rotate about their terminal hinge axis or *centric relation*)	A clinically reproducible and registerable position used in the dental rehabilitation of the edentulous patient as well as in the reorganized approach in the dentate patient
Lateral excursions (dynamic)	Occlusal relationship between the teeth when the mandible is shifted horizontally to the right or left Working side: the side to which the mandible has moved Non-working side: the side away from which the mandible has moved	Horizontal movement guided by working side: 1. palatal/labial surfaces of canines/incisors 2. cuspal inclines of premolars/molars 3. a combination of both. Horizontal movement may also be guided by non-working side contacts and condylar head–glenoid fossae relationships
Protrusion (dynamic)	The anterior/forward movement of the mandible with teeth in occlusion	Usually guided by the palatal contour of the maxillary incisors Leads to a posterior open bite—Christensen's phenomenon

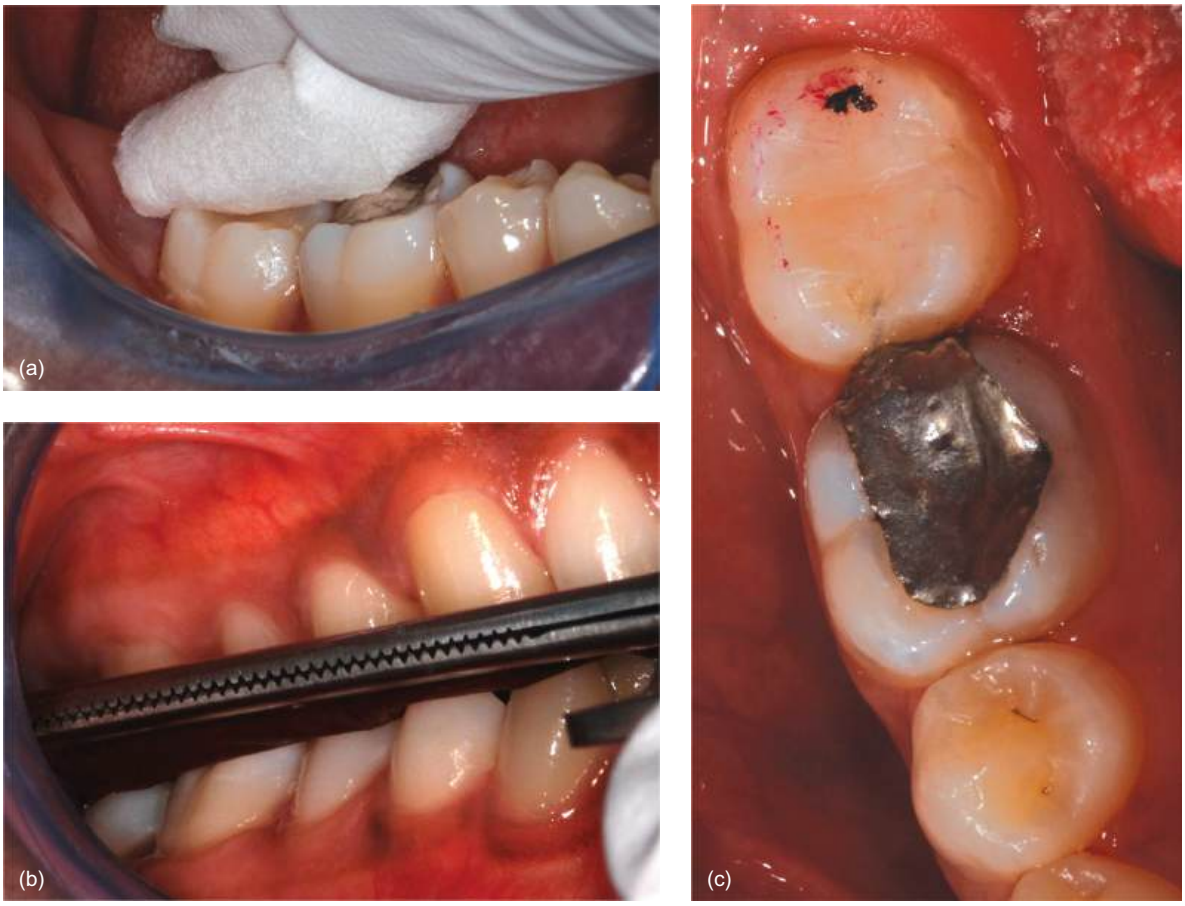


Figure 5.54 **(a)** Occlusal surfaces of LR quadrant being dried with a cotton wool roll. **(b)** Articulating paper held in forceps placed on occlusal surfaces and patient asked to bite together in the intercuspal position (ICP) and slide teeth left and right. **(c)** Heavy occlusal cusp–fossa contact shown on distal occlusal aspect of LR7 (red mark—ICP, black mark—lateral excursions). This can now be selectively removed with a rotary instrument.

Q5.12:

i Why is it important to dry the teeth prior to using dental articulating paper?

ii What mandibular movements should the patient shown in Figure 5.54 be encouraged to make when checking the occlusion using articulating paper?

- *Shimstock*: articulating foil (8 µm) for the fine occlusal adjustment of coronal restorations.
- *Clinical observation*: assess dental contacts around the arch prior to removing an old/damaged restoration, and ensure that these are replicated with the new restoration, especially in the partially dentate.

5.14.4 Clinical tips

- Occlusion must be checked both *before* and *after* placement of any restoration affecting the occlusal or guidance surfaces of teeth. Different coloured articulating paper can be used in this regard. This is essential in the conformative approach to dental care provision.
- Occlusal surfaces should be dried with cotton wool rolls (see Figure 5.54a).
- Occlusion must be checked dynamically as well as statically. The patient must be asked to 'grind' their teeth side to side and back to front in order to check that the guidance paths are free from interferences.
- Even though some authorities classify the buccal cusps of the mandibular teeth and palatal cusps of the maxillary dentition as *functional cusps*, it must be realized that all cusps, ridges, and fossae have an important role to play in a harmonious dynamic occlusion.

5.15 Suggested further reading and PubMed keywords

Department of Health (2009) *Delivering Better Oral Health. An evidence- based toolkit for prevention - second edition*. London: Department of Health/BASCD.
<www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsPolicyAndGuidance/DH_102331>

<www.ncbi.nlm.nih.gov/pubmed/advanced>

QR code image 5.3 Try searching the following keywords on PubMed for relevant further reading. You can access PubMed by scanning the QR code image or at the address www.ncbi.nlm.nih.gov/pubmed/advanced

Keywords

BDA infection control (UK); dental infection control; Department of Health (UK); HTM01-05; minimal intervention dentistry; minimally invasive dentistry; MI Compendium; glossary of prosthodontic terms.

5.16 Answers to self-test questions

Q5.1: Can you spot the gold crown in Figure 5.7?

A: It is present on the upper right-hand side of the radiograph—the discrete radio-opacity.

Q5.2: Can you spot an important clinical omission in Figure 5.17?

A: The safety floss around the connector arm and through the holes of the clamp is missing. This aids in retrieving pieces in cases of clamp separation.

Q5.3: In which clinical situations might you use each of the probes shown in Figure 5.21?

A: Straight probe—clinical examination of restoration margins and exposed carious dentine. Briault probe—proximal surfaces, internal cavity walls. Williams probe—periodontal pocket assessment, clinical examination of tooth surfaces. Naber's probe—periodontal furcation measurement. CPITN probe—periodontal assessment, clinical examination of tooth surfaces.

Q5.4: What are the advantages of having angulations between the working head and shank of the instruments pictured in Figure 5.24?

A: To enable the operator to access all aspects of a cavity, undercut, or undermined areas as well as offsetting the handle to allow better direct visual access to instrumentation.

Q5.5: What might be the advantages and disadvantages of single-use, disposable burs?

A: Advantages—improved infection control, no maintenance required. Disadvantages—potential long-term cost, possible concerns about quality control over the precision in manufacture, availability of all types of bur.

Q5.6: What clinical situations may benefit from using air-abrasion tooth preparation?

A: Minimally invasive dentistry procedures—diagnosing and treating carious fissures in high-risk patients with unmodifiable risk factors, tooth preparation for direct labial composite veneers, all tooth surface pre-treatment prior to adhesive bonding, stain removal/refreshing surfaces of old composites, potential selective enamel/dentine caries removal (with bioactive glass particles—in development).

Q5.7: Look at Figure 5.40. Why was the decision taken to retain such relatively large quantities of caries-affected dentine, especially at the gingival periphery of this particular lesion?

A: This was a conscious decision as an intact enamel periphery was present to create an adhesive seal. If the carious dentine was removed at the EDJ (as normal) this would result in a cavity margin that would

be located on root dentine at least 3–4 mm subgingivally, with no enamel present. Moisture control would then be impossible, no seal would be achievable, and the tooth/cavity would be unrestorable. By retaining this affected dentine, the tooth (dentine–pulp complex) has been given the opportunity to heal itself, with suitable input from the patient with regard to oral hygiene and dietary concerns.

Q5.8: Look at Figure 5.42. What is the name given to the appearance of the alternate light and dark striping evident in the inner third of enamel, arising from the EDJ and radiating towards the tooth surface?

A: Hunter–Schreger bands.

Q5.9: Which of the three levels do you think would be best to stop caries removal in the lesion shown in Figure 5.44?

A: Probably the second level (the green line) on caries-affected dentine where the collagen has partial structure to enable some form of dentine bonding to take place and the periphery histological status will allow a seal to be achieved, while not encroaching on the pulp and risking exposure.

Q5.10: Which hand instrument is shown in Figure 5.48?

A: Gingival margin trimmer.

Q5.11: After placing the direct pulp cap as shown in Figure 5.51, what are the next steps in the restoration of this cavity?

A: The setting calcium hydroxide should be protected with a layer of GIC/RMGIC. Then the adhesive restorative procedure can be performed—etch, bond, and placement of a resin composite restoration.

Q5.12:

i Why is it important to dry the teeth prior to using dental articulating paper?

A: To allow the ink from the paper to adhere to and mark the teeth in discrete areas of tooth–tooth contact.

ii What mandibular movements should the patient shown in Figure 5.54 be encouraged to make when checking the occlusion using articulating paper?

A: Vertical intercuspal position (the patient should be asked to bite up and down) and also lateral excursions (the patient should be asked to grind their teeth side to side to assess the cuspal inclines guiding the occlusion in these directions).